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(54) MODEL-BASED PROCESSING TO REDUCE REACTION TIMES FOR CONTENT STREAMING SYSTEMS AND APPLICATIONS

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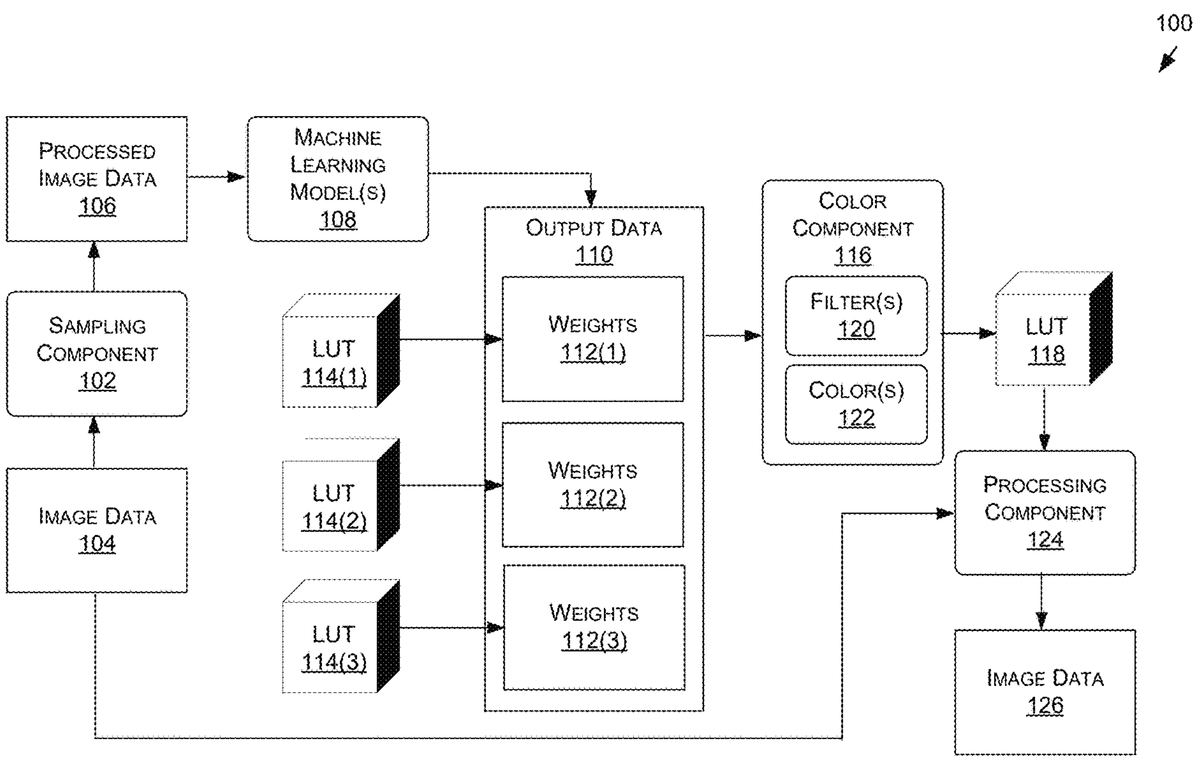
(57) **ABSTRACT**

In various examples, model-based processing to reduce reaction times for content streaming systems and applications is described herein. Systems and methods are disclosed that use one or more machine learning models to process image data representative of frames of an application, such as a gaming application, in order to generate updated imaged data representative of one or more updated frames that help reduce reaction times for users. For instance, the machine learning model(s) may update one or more visual characteristics associated with the frames, such as a contrast, a brightness, and/or a saturation associated with the frames. As described herein, the machine learning model(s) may be trained to update the frames in order to reduce the reaction times of users, such as by using one or more loss functions that measure loss in predicted reactions times and/or loss associated with visual characteristics of frames.



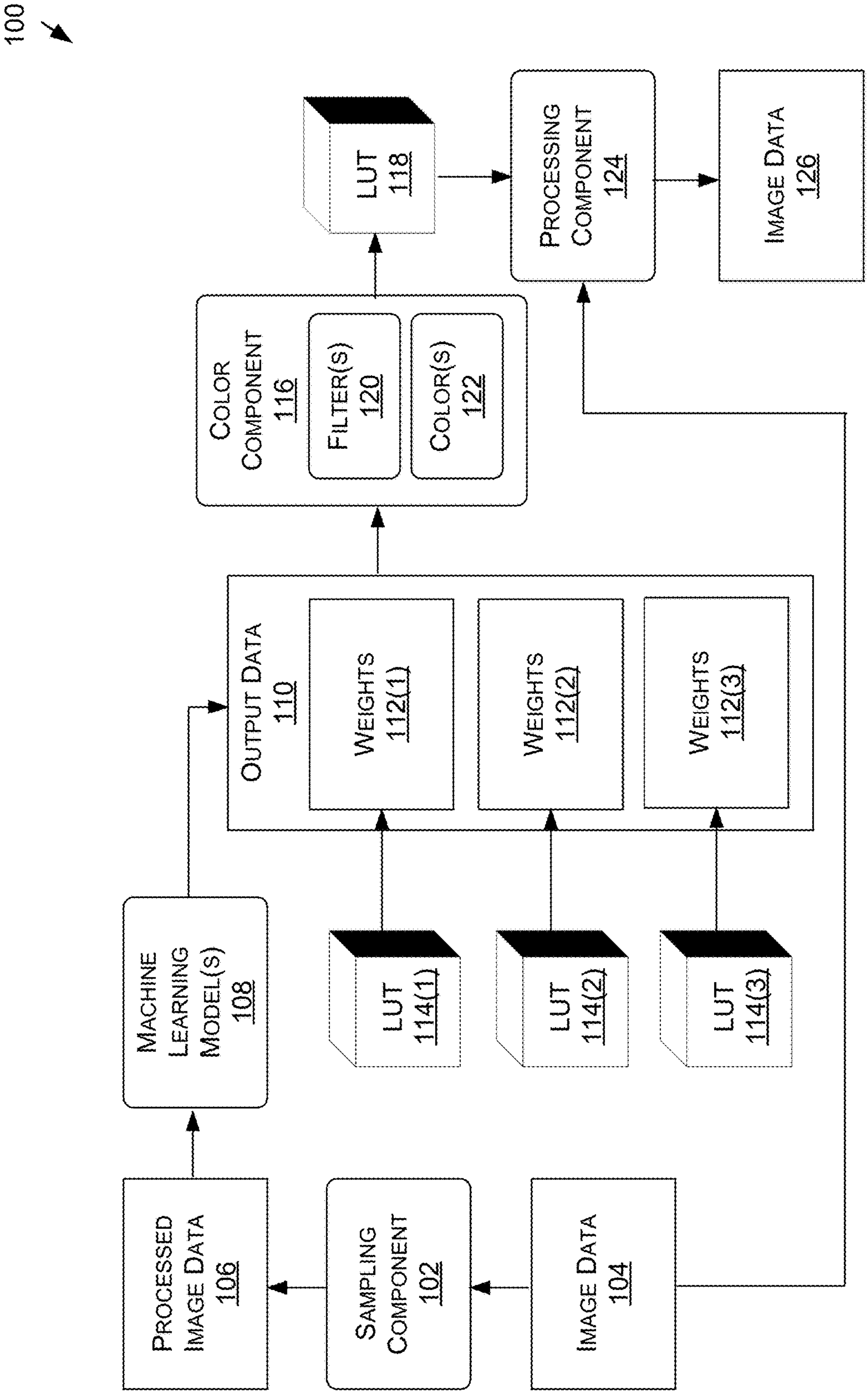

100
Processed Image Data 106
Sampling Component 102
Image Data 104
Machine Learning Model(s) 108
LUT 114(1)
LUT 114(2)
LUT 114(3)
Output Data 110
Weights 112(1)
Weights 112(2)
Weights 112(3)
Color Component 116
Filter(s) 120
Color(s) 122
LUT 118
Processing Component 124
Image Data 126

FIGURE 1A

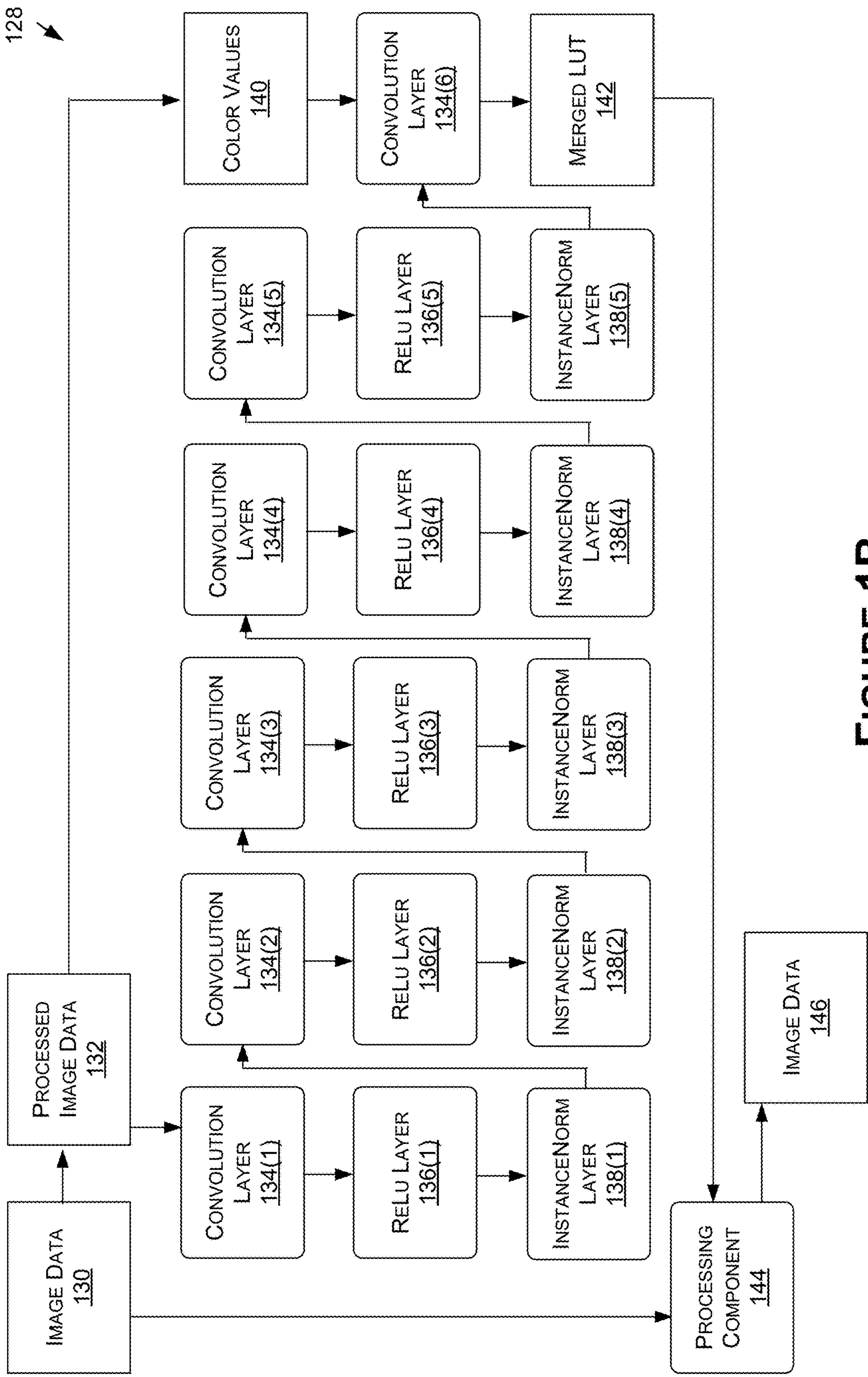

128
IMAGE DATA 130
PROCESSED IMAGE DATA 132
CONVOLUTION LAYER 134(1)
RELU LAYER 136(1)
INSTANCENORM LAYER 138(1)
CONVOLUTION LAYER 134(2)
RELU LAYER 136(2)
INSTANCENORM LAYER 138(2)
CONVOLUTION LAYER 134(3)
RELU LAYER 136(3)
INSTANCENORM LAYER 138(3)
CONVOLUTION LAYER 134(4)
RELU LAYER 136(4)
INSTANCENORM LAYER 138(4)
CONVOLUTION LAYER 134(5)
RELU LAYER 136(5)
INSTANCENORM LAYER 138(5)
COLOR VALUES 140
CONVOLUTION LAYER 134(6)
MERGED LUT 142
PROCESSING COMPONENT 144
IMAGE DATA 146

FIGURE 1B

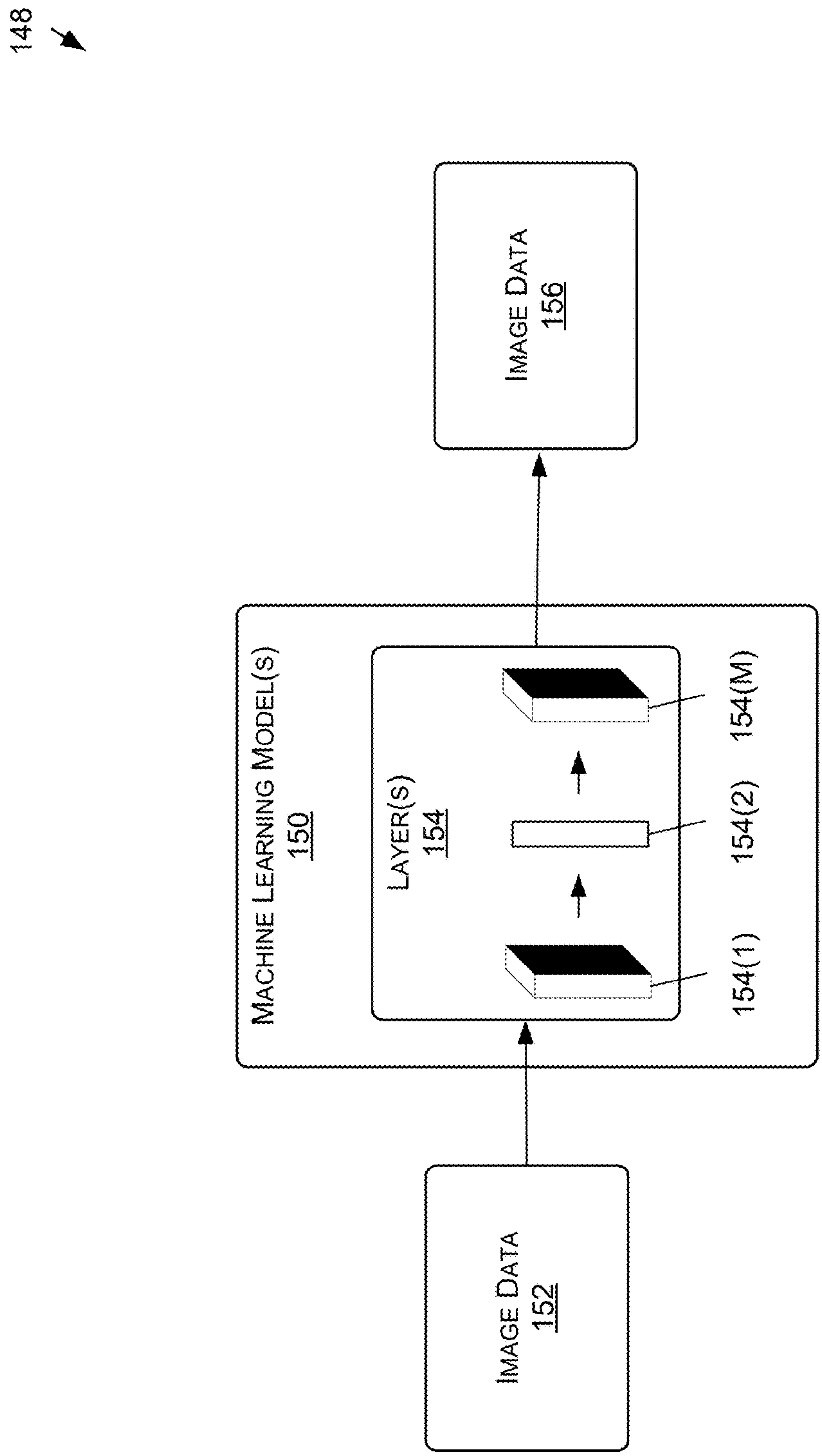

148
Image Data
152
Machine Learning Model(s)
150
Layer(s)
154
154(1)
154(2)
154(M)
Image Data
156

FIGURE 1C

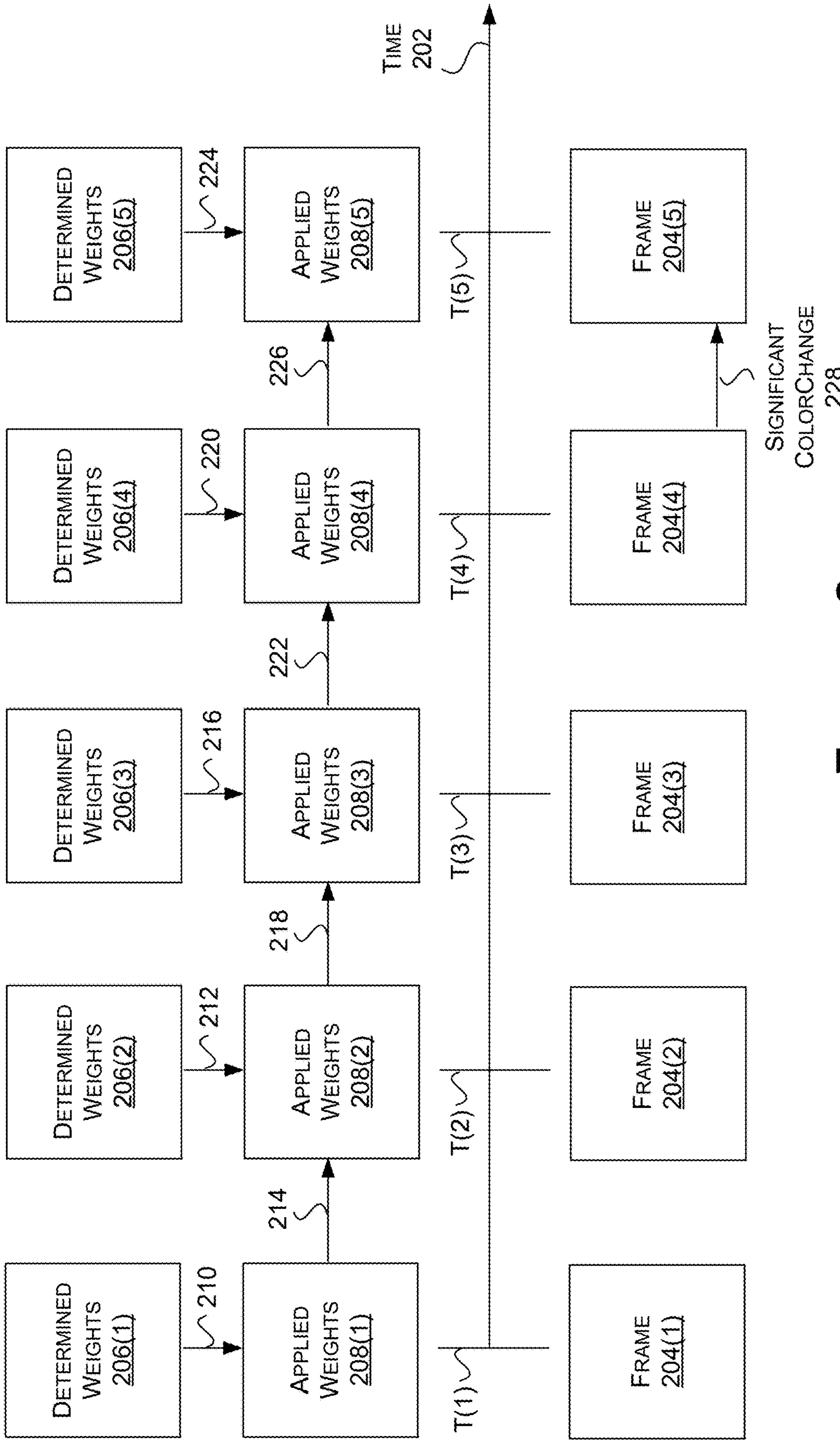

TIME
202
DETERMINED WEIGHTS 206(1)
DETERMINED WEIGHTS 206(2)
DETERMINED WEIGHTS 206(3)
DETERMINED WEIGHTS 206(4)
DETERMINED WEIGHTS 206(5)
210
212
216
220
224
APPLIED WEIGHTS 208(1)
APPLIED WEIGHTS 208(2)
APPLIED WEIGHTS 208(3)
APPLIED WEIGHTS 208(4)
APPLIED WEIGHTS 208(5)
214
218
222
226
T(1)
T(2)
T(3)
T(4)
T(5)
FRAME 204(1)
FRAME 204(2)
FRAME 204(3)
FRAME 204(4)
FRAME 204(5)
SIGNIFICANT
COLORCHANGE
228

FIGURE 2

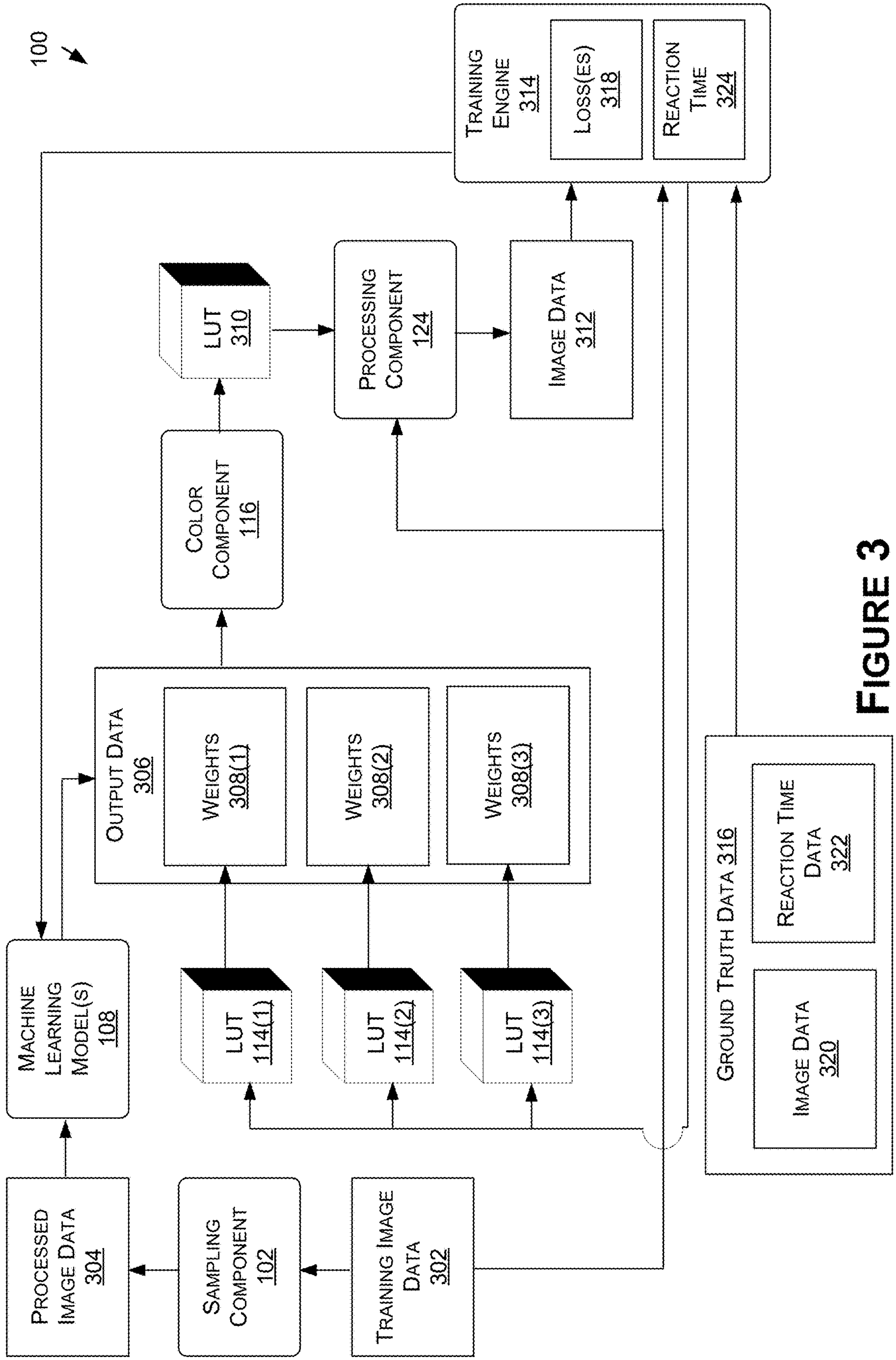

100
Processed Image Data 304
Sampling Component 102
Training Image Data 302
Machine Learning Model(s) 108
LUT 114(1)
LUT 114(2)
LUT 114(3)
Output Data 306
Weights 308(1)
Weights 308(2)
Weights 308(3)
Color Component 116
LUT 310
Processing Component 124
Image Data 312
Training Engine 314
Loss(es) 318
Reaction Time 324
Ground Truth Data 316
Image Data 320
Reaction Time Data 322

FIGURE 3

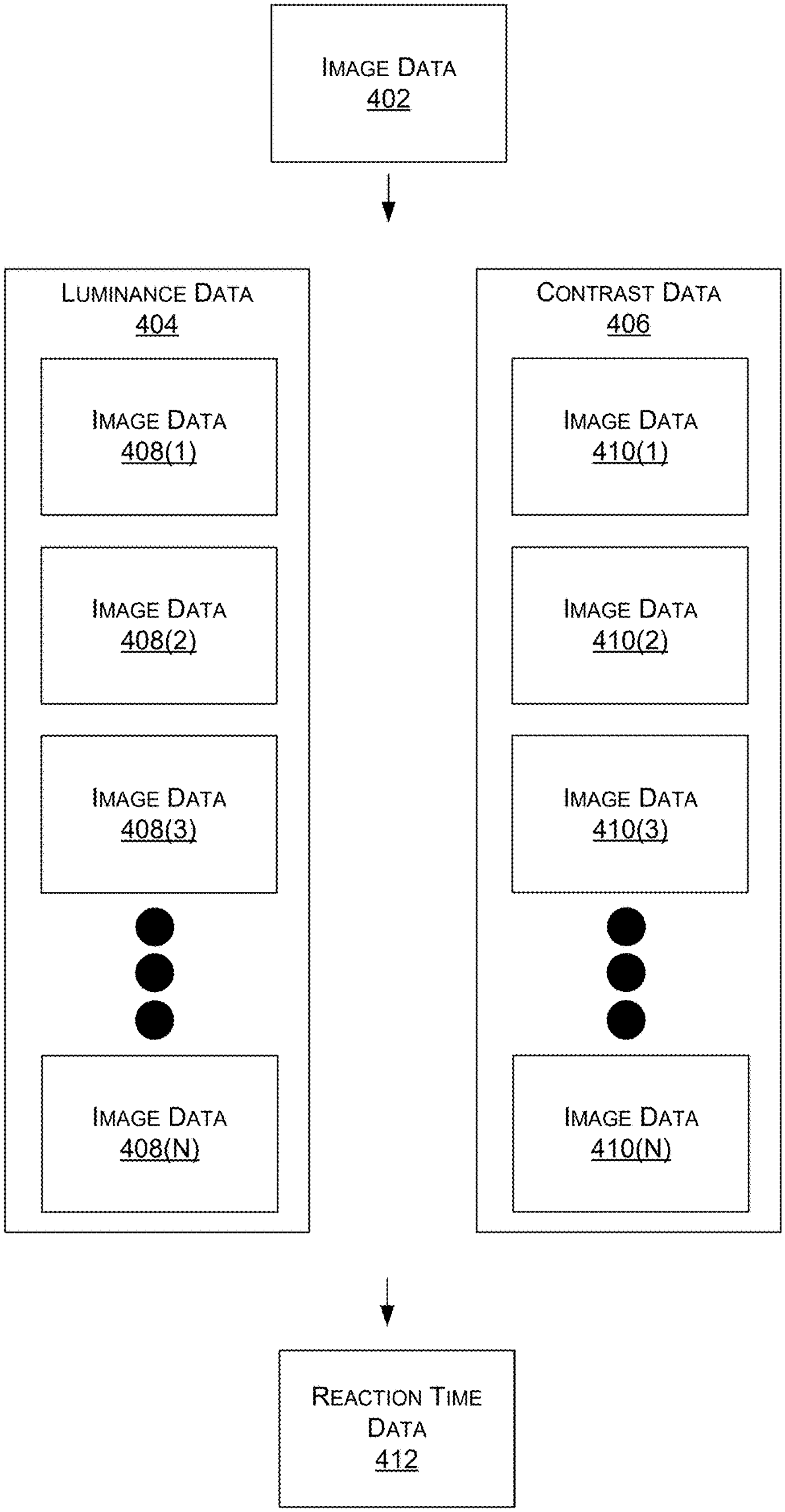


IMAGE DATA
402
LUMINANCE DATA
404
IMAGE DATA
408(1)
IMAGE DATA
408(2)
IMAGE DATA
408(3)
IMAGE DATA
408(N)
CONTRAST DATA
406
IMAGE DATA
410(1)
IMAGE DATA
410(2)
IMAGE DATA
410(3)
IMAGE DATA
410(N)
REACTION TIME
DATA
412

FIGURE 4

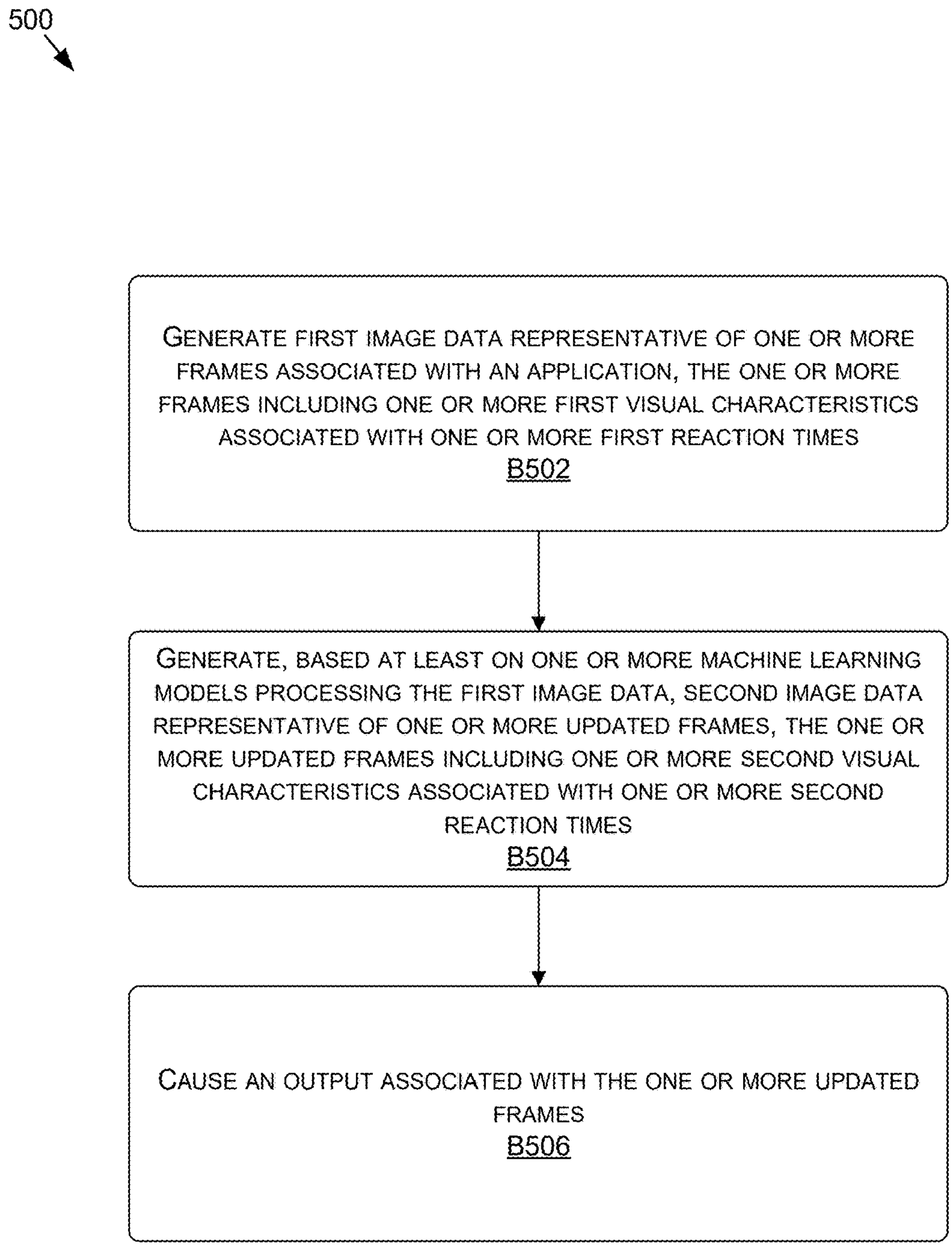

500
GENERATE FIRST IMAGE DATA REPRESENTATIVE OF ONE OR MORE FRAMES ASSOCIATED WITH AN APPLICATION, THE ONE OR MORE FRAMES INCLUDING ONE OR MORE FIRST VISUAL CHARACTERISTICS ASSOCIATED WITH ONE OR MORE FIRST REACTION TIMES
B502
GENERATE, BASED AT LEAST ON ONE OR MORE MACHINE LEARNING MODELS PROCESSING THE FIRST IMAGE DATA, SECOND IMAGE DATA REPRESENTATIVE OF ONE OR MORE UPDATED FRAMES, THE ONE OR MORE UPDATED FRAMES INCLUDING ONE OR MORE SECOND VISUAL CHARACTERISTICS ASSOCIATED WITH ONE OR MORE SECOND REACTION TIMES
B504
CAUSE AN OUTPUT ASSOCIATED WITH THE ONE OR MORE UPDATED FRAMES
B506

FIGURE 5

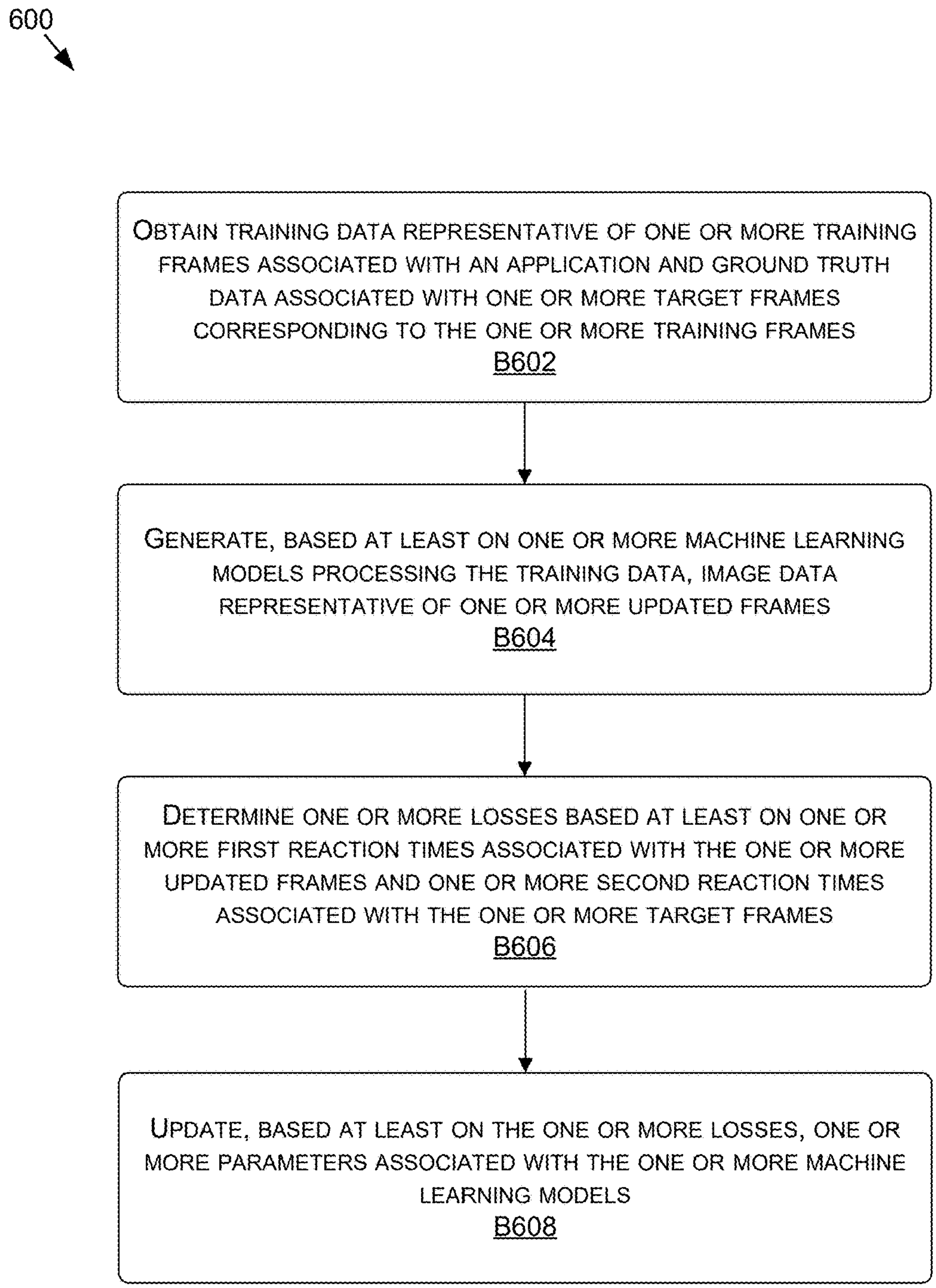

600
OBTAIN TRAINING DATA REPRESENTATIVE OF ONE OR MORE TRAINING FRAMES ASSOCIATED WITH AN APPLICATION AND GROUND TRUTH DATA ASSOCIATED WITH ONE OR MORE TARGET FRAMES CORRESPONDING TO THE ONE OR MORE TRAINING FRAMES
B602
GENERATE, BASED AT LEAST ON ONE OR MORE MACHINE LEARNING MODELS PROCESSING THE TRAINING DATA, IMAGE DATA REPRESENTATIVE OF ONE OR MORE UPDATED FRAMES
B604
DETERMINE ONE OR MORE LOSSES BASED AT LEAST ON ONE OR MORE FIRST REACTION TIMES ASSOCIATED WITH THE ONE OR MORE UPDATED FRAMES AND ONE OR MORE SECOND REACTION TIMES ASSOCIATED WITH THE ONE OR MORE TARGET FRAMES
B606
UPDATE, BASED AT LEAST ON THE ONE OR MORE LOSSES, ONE OR MORE PARAMETERS ASSOCIATED WITH THE ONE OR MORE MACHINE LEARNING MODELS
B608

FIGURE 6

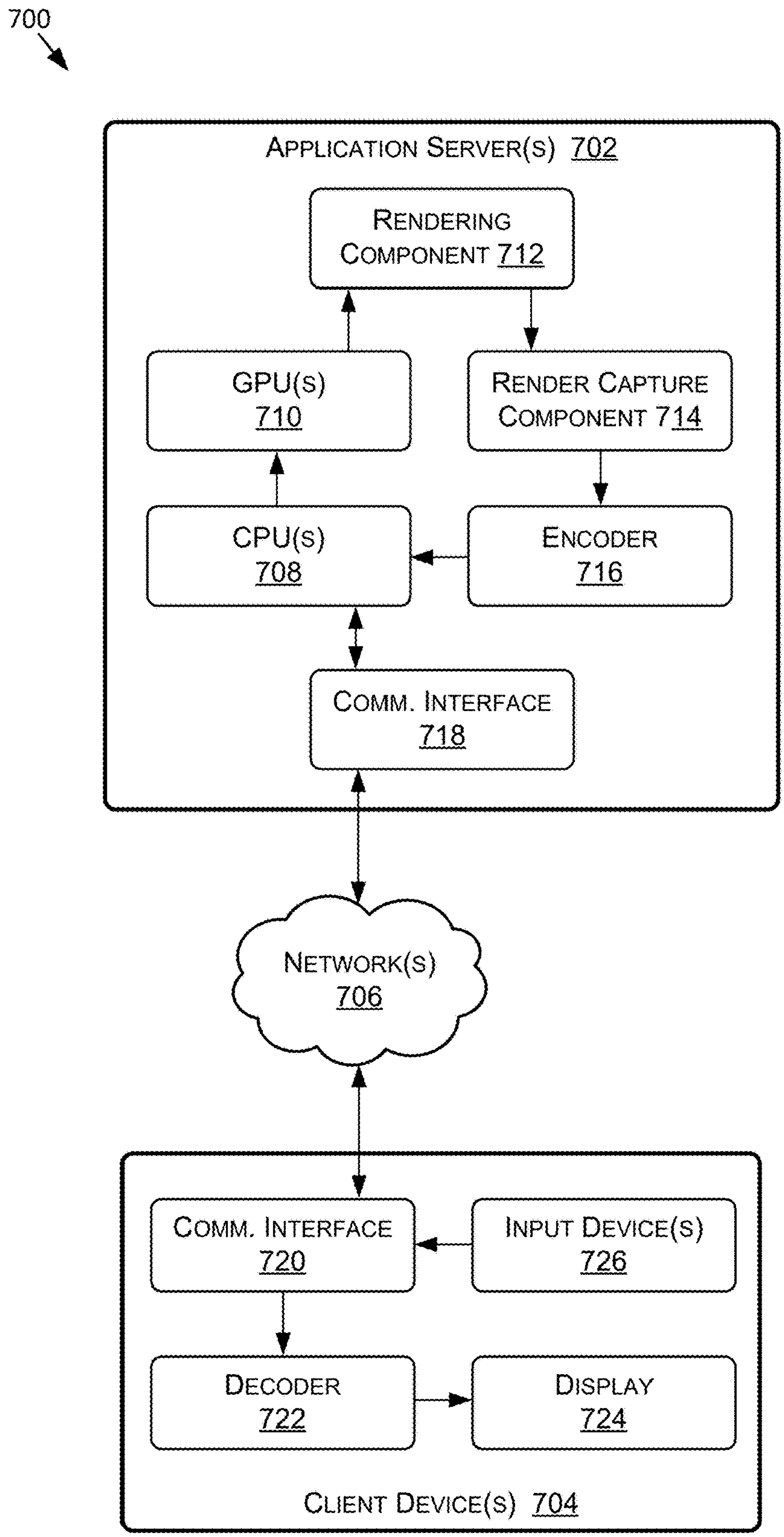

700
Application Server(s) 702
Rendering Component 712
GPU(s) 710
Render Capture Component 714
CPU(s) 708
Encoder 716
Comm. Interface 718
Network(s) 706
Comm. Interface 720
Input Device(s) 726
Decoder 722
Display 724
Client Device(s) 704

FIGURE 7

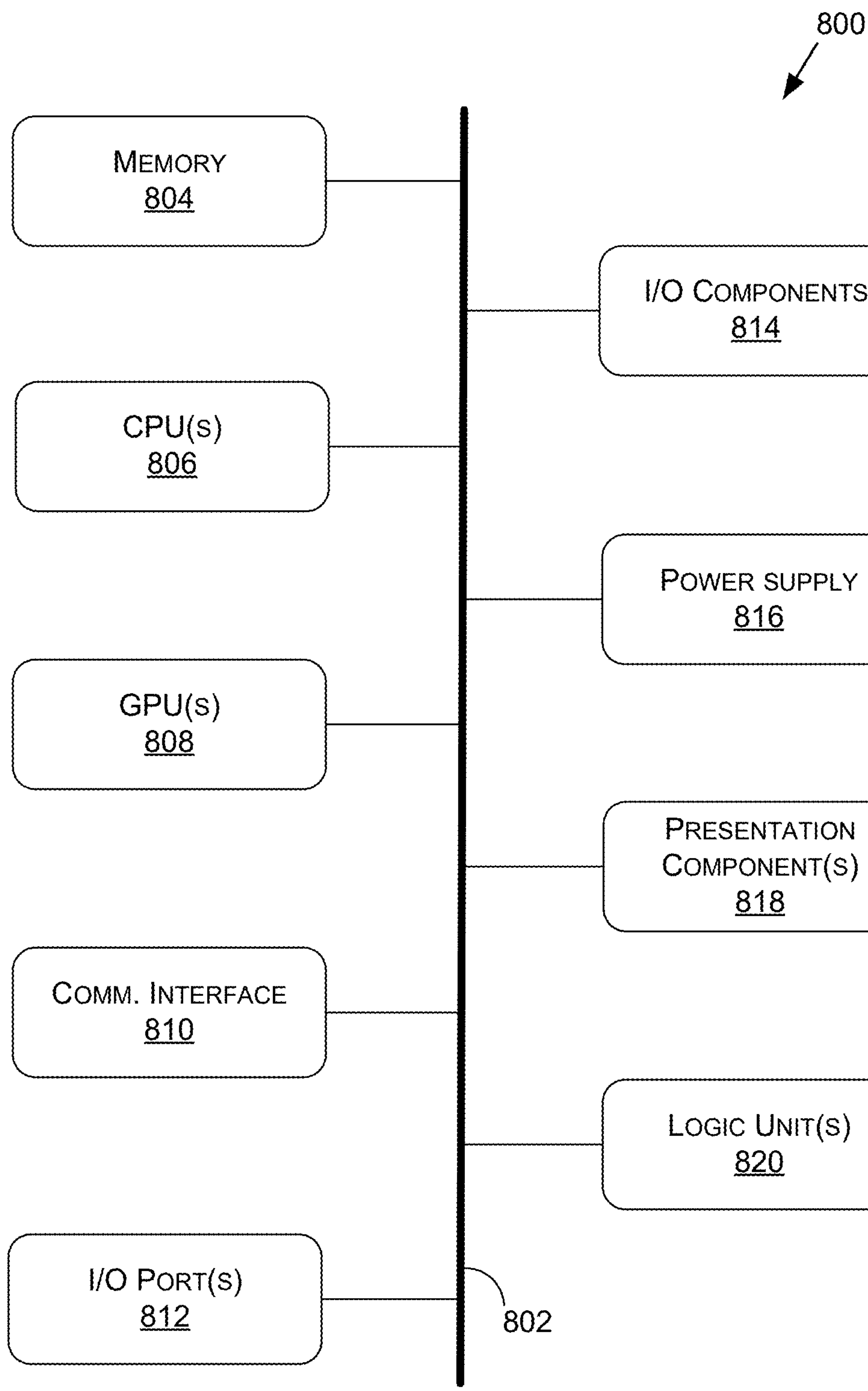

800
MEMORY
804
I/O COMPONENTS
814
CPU(S)
806
POWER SUPPLY
816
GPU(S)
808
PRESENTATION
COMPONENT(S)
818
COMM. INTERFACE
810
LOGIC UNIT(S)
820
I/O PORT(S)
812
802

FIGURE 8

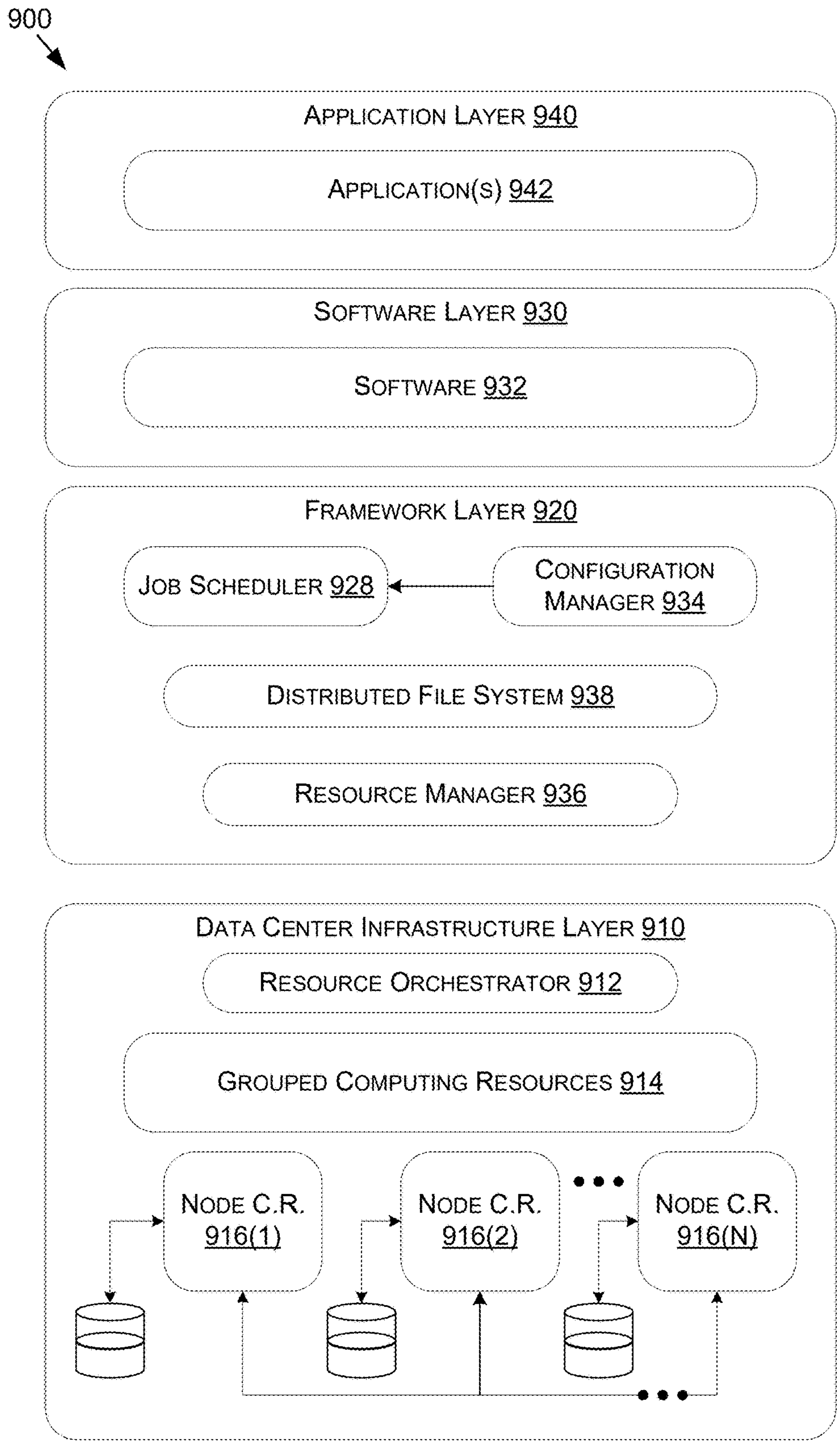

900
APPLICATION LAYER 940
APPLICATION(S) 942
SOFTWARE LAYER 930
SOFTWARE 932
FRAMEWORK LAYER 920
JOB SCHEDULER 928
CONFIGURATION MANAGER 934
DISTRIBUTED FILE SYSTEM 938
RESOURCE MANAGER 936
DATA CENTER INFRASTRUCTURE LAYER 910
RESOURCE ORCHESTRATOR 912
GROUPED COMPUTING RESOURCES 914
NODE C.R. 916(1)
NODE C.R. 916(2)
NODE C.R. 916(N)

FIGURE 9

MODEL-BASED PROCESSING TO REDUCE REACTION TIMES FOR CONTENT STREAMING SYSTEMS AND APPLICATIONS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/556,397 filed on Feb. 22, 2024, which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] For some gaming applications, the success of a player may be based on the player's reaction time, such as the time it takes for the player to observe what events are occurring with regard to a gaming application, process the events to determine how to respond, and then react with the determined response. Currently, there are many factors that may contribute to the player's reaction time, such as visual characteristics associated with frames being presented to the player. For instance, the contrast, brightness, saturation, and/or other visual characteristics of the frames may affect the player's reaction time. For example, if the visual characteristics include higher contrast levels, brightness levels, and/or saturation levels, then the player's reaction time may decrease as the player may more easily observe and process the events. However, if the visual characteristics include lower contrast levels, brightness levels, and/or saturation levels, then the player's reaction time may increase as it may be more difficult for the player to observe and/or process the events.

[0003] As such, some conventional systems may allow for players to set levels for various visual characteristics associated with gaming applications. For instance, a player may set a contrast level, a brightness level, and/or a saturation level that the conventional systems may then use when processing frames of a gaming application. For example, if the player increases the contrast level, the brightness level, and/or the saturation level, then the conventional systems may update the visual characteristics of the frames by increasing the contrast levels, the brightness levels, and/or the saturation levels associated with the frames. However, in some circumstances, while applying updates to visual characteristics for some frames may decrease the reaction times for players, the same static updates to visual characteristics of other frames may have the opposite effect of increasing reaction times. This is because the same static updates to visual characteristics may cause some events to be easier to observe in some frames while also causing other events to be more difficult to observe in other frames.

SUMMARY

[0004] Embodiments of the present disclosure relate to model-based processing to reduce reaction times for content streaming systems and applications. Systems and methods are disclosed that use one or more machine learning models to process image data representative of frames of an application, such as a gaming application, in order to generate updated image data representative of one or more updated frames that help reduce reaction times for users. For instance, based at least on the processing, the machine learning model(s) may update one or more visual characteristics associated with the frames, such as contrast levels, brightness levels, and/or saturation levels associated with the frames. As described in more detail herein, the machine learning model(s) may be trained to update the frames in order to reduce the reaction times of users, such as by using one or more loss functions that measure loss in predicted reaction times and/or loss associated with visual characteristics of frames.

[0005] In contrast to conventional systems, such as those described above, the system of the present disclosure may dynamically update levels of various characteristics that are applied to frames using the machine learning model(s). For instance, and as discussed above, the conventional systems may apply the same, static levels of visual characteristics to all frames, which may decrease the reaction times for some frames, but may also increase the reaction times for other frames. In contrast, by dynamically updating the levels of visual characteristics using the machine learning model(s), individual frames and/or groups of frames may be updated using different levels of visual characteristics, such as respective contrast levels, brightness levels, and/or saturation levels, that are configured to reduce (e.g., minimize) the reaction times for at least a majority (e.g., all) of the frames.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The present systems and methods for model-based processing to reduce reaction times for content streaming systems and applications are described in detail below with reference to the attached drawing figures, wherein:

[0007] FIGS. **1A-1C** illustrate examples of processes of using one or more machine learning models to process frames in order to reduce reaction times associated with an application, in accordance with some embodiments of the present disclosure;

[0008] FIG. **2** illustrates an example of using a filter to determine weights associated with frames, in accordance with some embodiments of the present disclosure;

[0009] FIG. **3** illustrates an example of a process of training one or more machine learning models to process frames in order to reduce reaction times associated with an application, in accordance with some embodiments of the present disclosure;

[0010] FIG. **4** illustrates an example of determining reaction times associated with a frame, in accordance with some embodiments of the present disclosure;

[0011] FIG. **5** illustrates a flow diagram showing a method for using one or more machine learning models to process frames in order to reduce reaction times associated with an application, in accordance with some embodiments of the present disclosure;

[0012] FIG. **6** illustrates a flow diagram showing a method for training one or more machine learning models to process frames in order to reduce reaction times associated with an application, in accordance with some embodiments of the present disclosure;

[0013] FIG. **7** is a block diagram of an example content streaming system suitable for use in implementing some embodiments of the present disclosure;

[0014] FIG. **8** is a block diagram of an example computing device suitable for use in implementing some embodiments of the present disclosure; and

[0015] FIG. **9** is a block diagram of an example data center suitable for use in implementing some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0016] Systems and methods are disclosed related to model-based processing to reduce reaction times for content streaming systems and applications. For instance, a system (s) may host a session of an application with a client device. As described herein, an application may include, but is not limited to, a gaming application, an interactive application, a multimedia application (e.g., a video streaming application, a music streaming application, a voice streaming application, a multimedia streaming application that includes both audio and video, etc.), a communications application (e.g., a video conferencing application, etc.), an educational application, a collaborative content creation application, or any other type of application. For instance, during the session, the client device may generate input data representing one or more inputs received from the user via one or more input devices. The system(s) may then receive the input data from the user, use the input data to update a state of the application, and then send content data (e.g., image data, audio data, etc.) representing the current state of the application back to the client device. Using the content data, the client device may then provide the content to the user, such as by displaying one or more frames depicting the current state and/or outputting sound associated with the current state. This process may then continue to repeat during the session associated with the application.

[0017] As described herein, the system(s) may process the image data using one or more machine learning models that are trained to update frames in order to reduce reaction times associated with users of applications. For instance, and for a frame, the system(s) and/or the machine learning model(s) may downsample the frame in order to generate a processed frame that includes a lower resolution than the original frame. The machine learning model(s) may then process the processed frame using one or more trained layers and, based at least on the processing, determine one or more weights associated with one or more initial lookup tables (LUT). In some examples, the machine learning model(s) may be configured to determine any number of weights associated with any number of initial LUTs, such as one weight associated with one initial LUT, two weights associated with two initial LUTs, three weights associated with three initial LUTs, five weights associated with five initial LUTs, and/or so forth. In some examples, the system(s) and/or the machine learning model(s) may then generate a final LUT by combining (e.g., concatenating, adding, fusing, etc.) the weight(s) associated with the initial LUT(s). The system(s) and/or the machine learning model(s) may apply the final LUT to the frame in order to generate an updated frame.

[0018] Additionally, or alternatively, in some examples, the system(s) and/or the machine learning model(s) may directly apply the initial LUT(s) to the frame in order to generate the updated frame. In such examples, the initial LUT(s) may be applied to the frame based at least on the weight(s) determined to be associated with the LUT(s). In either of the examples, by performing such processes, the final frame may include one or more values associated with one or more visual characteristics (e.g., a contrast, a brightness, a saturation, etc.) that cause the updated frame to be associated with a reaction time that is less than a reaction time associated with the original frame, which is described in more detail herein.

[0019] The system(s) may then continue to perform these processes of using the machine learning model(s) to process new frames in order to generate updated frames associated with the application. In some examples, since the system(s) process individual frames and/or groups of frames separately using the machine learning model(s), the updates applied by the machine learning model(s) to the frames may be dynamic such that the individual frames and/or the groups of frames include visual characteristics for reducing the reaction time of the user. Additionally, in some examples, and also based on processing the individual frames and/or the groups of frames separately, the machine learning model(s) may include one or more filters that are configured to ensure smoothness and/or consistency (e.g., temporal consistency) to the updates across the frames.

[0020] For instance, while updating the frames, the system (s) and/or the machine learning model(s) may further process the frames in order to determine color values (e.g., brightness values, etc.) associated with the frames, such as mean color values, mode color values, minimum color values, maximum color values, and/or any other color value. The system(s) and/or the machine learning model(s) may then determine whether there is a significant change in color between two consecutive claims using the color values. In some examples, the system(s) and/or the machine learning model(s) may determine whether there is a significant change based on tracking a mean color value between frames while performing the updating. For example, the system(s) and/or the machine learning model(s) may determine that there is a significant change when the color value associated with a frame differs from the mean color value by a threshold amount and/or determine that there is not a significant change when the color value does not differ from the mean color value by the threshold amount. While this is just one example technique of how the system(s) and/or the machine learning model(s) may determine when there is a significant change in color between consecutive frames, in other examples, the system(s) and/or the machine learning model(s) may use one or more additional and/or alternative techniques.

[0021] The machine learning model(s) may then use the determinations of whether there are significant changes in color when updating frames. For instance, and as described above, the machine learning model(s) may determine the weight(s) associated with the initial LUT(s). As such, in some examples, the machine learning model(s) may filter the weights over a period of time and/or over a number of frames, such as by continuing to update the weight(s) as new frames are received. For example, the machine learning model(s) may determine a first weight(s) associated with a first frame, determine a second weight(s) associated with a second frame by updating the first weight(s) using a new weight(s) determined for the second frame, determine a third weight(s) associated with a third frame by updating the second weight(s) using a new weight(s) determined for the third frame, and/or so forth. In other words, the machine learning model(s) may filter the weights associated with the frames such that the weights remain smooth between the frames (e.g., there are no large changes in colors between the frames).

[0022] As such, in some examples, if there is no significant change in color between two consecutive frames, then the system(s) and/or the machine learning model(s) may determine the filtered weight(s) associated with the subsequent frame by providing more weight (e.g., 90%) to the previously filtered weight(s) than the new weight(s) determined for the subsequent frame. In other words, if there is no significant change in color between two consecutive frames, then the system(s) and/or the machine learning model(s) may not change the weights for the frames by a large amount. However, if there is significant change in color between two consecutive frames, then the system(s) and/or the machine learning model(s) may determine the filtered weight(s) associated with the subsequent frame by providing less weight (e.g., 10%) to the previously filtered weight(s) as compared to the new weight(s) determined for the frame. In other words, if there is a significant change in color between two consecutive frames, then the system(s) and/or the machine learning model(s) may change the weights for the frames by a large amount. This way, when updating frames using one or more of the processes described herein, frames that are close in color with one another (e.g., there is no significant change) may include similar color characteristic levels applied during updating while frames that are not close in color with another (e.g., there is a significant change in color) may include different color characteristic levels applied.

[0023] As described herein, in some examples, the system (s) may train (e.g., update, etc.) the machine learning model (s) using one or more techniques such that the machine learning model(s) is able to update frames in a way that reduces reaction times of users. For instance, and for training data representing a training frame, the system(s) may apply the training frame to the machine learning model(s). The system(s) and/or the machine learning model(s) may then process the training frame using one or more of the processes described herein and, based at least on the processing, generate an updated frame corresponding to the training frame. The system(s) may then determine one or more losses using the training frame and the updated frame. For instance, in some examples, the system(s) may determine one or more first losses based at least on comparing the updated frame to a target frame (which may be represented by ground truth data) that is associated with the training frame. As will be described in more detail herein, the target frame may be similar to the training frame, but with being updated using one or more visual characteristics. For instance, the target frame may correspond to the training frame that is updated to include greater contrast, greater brightness, and/or greater saturation.

[0024] Additionally, or alternatively, in some examples, the system(s) may determine one or more second losses based at least on reaction times associated with the updated frame and the target frame. For instance, and as will be described in more detail herein, the system(s) may process the updated frame and, based at least on the processing, compute one or more first reaction time predictions associated with one or more points (e.g., one or more pixels) of the updated frame. Additionally, the system(s) may process the target frame and, based at least on the processing, compute one or more second reaction time predictions associated with the one or more points (e.g., one or more pixels) of the target frame (and/or the ground truth data may represent the second reaction time prediction(s)). The system(s) may then determine the second loss(es) based at least on the first reaction time prediction(s) and the second reaction time prediction(s).

[0025] The system(s) may then determine one or more total losses associated with the training frame using the first loss(es) and/or the second loss(es). Additionally, the system (s) may use the total loss(es) to update the machine learning model(s). For instance, in some examples, the system(s) may use the total loss(es) to update one or more parameters and/or weights associated with the layer(s) of the machine learning model(s) that is trained to determine the weight(s) associated with the initial LUT(s). Additionally, or alternatively, in some examples, the system(s) may use the total loss(es) to update the initial LUT(s) associated with the machine learning model(s). In any of the examples, the system(s) may then continue to perform these processes using one or more additional training frames and/or ground truth data corresponding to the additional training frame(s).

[0026] The systems and methods described herein may be used by, without limitation, non-autonomous vehicles or machines, semi-autonomous vehicles or machines (e.g., in one or more adaptive driver assistance systems (ADAS)), autonomous vehicles or machines, piloted and un-piloted robots or robotic platforms, warehouse vehicles, off-road vehicles, vehicles coupled to one or more trailers, flying vessels, boats, shuttles, emergency response vehicles, motorcycles, electric or motorized bicycles, aircraft, construction vehicles, underwater craft, drones, and/or other vehicle types. Further, the systems and methods described herein may be used for a variety of purposes, by way of example and without limitation, for machine control, machine locomotion, machine driving, synthetic data generation, model training, perception, augmented reality, virtual reality, mixed reality, robotics, security and surveillance, simulation and digital twinning, autonomous or semi-autonomous machine applications, deep learning, environment simulation, object or actor simulation and/or digital twinning, data center processing, conversational AI, light transport simulation (e.g., ray-tracing, path tracing, etc.), collaborative content creation for 3D assets, cloud computing and/or any other suitable applications.

[0027] Disclosed embodiments may be comprised in a variety of different systems such as automotive systems (e.g., a control system for an autonomous or semi-autonomous machine, a perception system for an autonomous or semi-autonomous machine), systems implemented using a robot, aerial systems, medial systems, boating systems, smart area monitoring systems, systems for performing deep learning operations, systems for performing simulation operations, systems for performing digital twin operations, systems implemented using an edge device, systems implementing large language models (LLMs), systems implementing vision language models (VLMs), systems incorporating one or more virtual machines (VMs), systems for performing synthetic data generation operations, systems implemented at least partially in a data center, systems for performing conversational AI operations, systems for performing light transport simulation, systems for performing collaborative content creation for 3D assets, systems for performing generative AI operations, systems implemented at least partially using cloud computing resources, and/or other types of systems.

[0028] With reference to FIG. **1**A, FIG. **1**A illustrates an example of a process **100** of using one or more machine learning models to process frames in order to reduce reaction times associated with an application, in accordance with some embodiments of the present disclosure. It should be understood that this and other arrangements described herein are set forth only as examples. Other arrangements and elements (e.g., machines, interfaces, functions, orders, groupings of functions, etc.) may be used in addition to or instead of those shown, and some elements may be omitted altogether. Further, many of the elements described herein are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, and in any suitable combination and location. Various functions described herein as being performed by entities may be carried out by hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory.

[0029] The process **100** may include a sampling component **102** receiving image data **104**, where the image data **104** represents one or more frames associated with an application. For instance, one or more application servers (e.g., an application server(s) **702**) may be providing the application to one or more client devices (e.g., a client device **704**). As described herein, the application may include, but is not limited to, a gaming application, an interactive application, a multimedia application (e.g., a video streaming application, a music streaming application, a voice streaming application, a multimedia streaming application that includes both audio and video, etc.), a communications application (e.g., a video conferencing application, etc.), an educational application, a collaborative content creation application, or any other type of application. As such, the application server(s) may generate and then send the image data **104** for output by the client device. However, in order to help reduce the reaction times of one or more users associated with the application, the application server (s) and/or the client device may process the image data **104**.

[0030] For instance, the process **100** may include the sampling component **102** processing the image data **104** in order to generate processed image data **106**. In some examples, the sampling component **102** may process the image data **104** using one or more techniques, such as by downsampling one or more frames represented by the image data **104**. For example, the frame(s) represented by the image data **104** may include a first resolution and the frame(s) represented by the processed image data **106** may include a second, lower resolution. As described herein, a frame resolution may include, but is not limited to, 256×256, 1920×1080, 2960×1440, 3840×2160, and/or any other frame resolution.

[0031] The process **100** may then include applying the processed image data **106** to one or more machine learning models **108**. As described herein, the machine learning model(s) **108** may include, but is not limited to, one or more convolutional neural networks, one or more probabilistic neural networks, one or more feedforward neural networks, one or more deep stacking networks, and/or any other type of network. As shown, the process **100** may then include the machine learning model(s) **108** processing the processed image data **106** and, based at least on the processing, generating output data **110** associated with the image data **104**. As shown, the output data **110** may represent weights **112(1)**-(**3**) (also referred to singularly as "weight **112**" or in plural as "weights **112**") associated with LUTs **114(1)**-(**3**) (also referred to singularly as "LUT **114**" or in plural as "LUTs **114**"). While the example of FIG. **1**A illustrates the machine learning model(s) **108** determining three weights **112** for three different LUTs **114**, in other examples, the machine learning model(s) **108** may determine any number of weights associated with any number of LUTs.

[0032] As described herein, a LUT **114** may include a lookup table, such as a one-dimensional (1D) lookup table, two-dimensional (2D) lookup table, a three-dimensional (3D) lookup table, and/or any other type of lookup table that is used to transform one or more visual characteristics associated with the frame(s) represented by the image data **104**. As described herein, a visual characteristic may include, but is not limited to, a contrast, a brightness, a saturation, a color, a sharpness, a tone, and/or any other visual characteristic. In some examples, the machine learning model(s) **108** and/or the LUTs **114** may be used to process image data **104** generated for a specific application. In some examples, the machine learning model(s) **108** and/or the LUTs may be used to process image data **104** generated for multiple applications. Still, in some examples, and as described more herein, the machine learning model(s) **108** and/or the LUTs **114** may be generated and/or updated during a training process associated with the machine learning model(s) **108**.

[0033] For instance, and as shown, the process **100** may include a color component **116** using the weights **112** and the LUTs **114** to generate a LUT **118** that is used to then process the image data **104**. In some examples, and for a frame, the color component **116** may generate the LUT **118** by combining (e.g., concatenating, adding, fusing, etc.) the LUTs **114** based at least on the weights **112**. For example, the color component **116** may apply the first weight **112**(**1**) to the first LUT **114**(**1**) to generate a first output, apply the second weight **112**(**2**) to the second LUT **114**(**2**) to generate a second output, and also apply the third weight **112**(**3**) to the third LUT **114**(**3**) to generate a third output. In some examples, the color component **116** may then fuse (e.g., add, multiply, interpolate, and/or perform any other operation) the outputs in order to generate the LUT **118**. As described herein, in some examples, since the machine learning model (s) **108** determines the weights **112** based on reducing reaction times, by performing these processes, the LUT **118** may thus be configured to update visual characteristics of the frame such that the frame is associated with a lower reaction time.

[0034] In some examples, and based at least on processing the individual frames and/or the groups of frames separately, the color component **116** may include one or more filters **120** that are configured to ensure smoothness, compatibility, and/or so forth to the updates across the frames. For instance, the color component **116** may process the frames represented by the image data **104** (and/or the processed image data **106**) using a first filter **120** in order to determine color values **122** associated with the frames, such as brightness values. For example, the first filter **120** may track the mean color values **122**, the mode color values **122**, the minimum color values **122**, the maximum color values **122**, and/or any other type of color values **122** across the frames. As described herein, the first filter **120** may track the color values over a specific number of frames (e.g., one frame, ten frames, fifty frames, etc.) using a sliding window, over a specific period of time (e.g., one second, five seconds, ten seconds, etc.) using a sliding widow, continuously until detecting a significant change in color, and/or using any other technique.

[0035] The color component **116** may then use the first filter **120** to determine when there are significant changes in color associated with frames. For example, the color component **116** may determine that there is a significant change in color between two consecutive frames based at least on the color value **122** associated with the subsequent frame differing from the current mean color value **122** by a threshold amount, or determine that there is no significant change between the two consecutive frames based at least on the color value **122** associated with the subsequent frame being within the threshold amount to the current mean color value **122**. While this is just one example technique of how the color component **116** may determine when there is a significant change in color between consecutive frames, in other examples, the color component **116** may use one or more additional and/or alternative techniques.

[0036] The color component **116** may then use the determinations of whether there are significant changes in color when updating frames. For instance, and for a current frame represented by the image data **104**, the color component **116** may use a second filter **120** to determine final weights **112** to apply to the LUTs **114** using one or more previous weights **112** for one or more previous frames and current weights **112** determined for the current frame. As described herein, the second filter **120** may determine the weights **112** over a specific number of frames (e.g., one frame, ten frames, fifty frames, etc.) using a sliding window, over a specific period of time (e.g., one second, five seconds, ten seconds, etc.) using a sliding widow, continuously until detecting a significant change in color, and/or using any other technique.

[0037] For instance, the second filter **120** may blend the current weights **112** associated with the current frame with the previous weights **112** associated with the previous frames to determine final weights **112**. When performing the blending, the second filter **120** may apply a greater weight to the previous weights **112** as compared to the current weights **112** when there is not a significant change in color associated with the current frame. For example, the second filter **120** may apply 90% (and/or any other percentage) to the previous weights **112** and 10% (and/or any other percentage) to the current weights **112** when there is no significant change in color. This way, the color processing performed on the current frame may be similar to the color processing performed on the previous frame in order to ensure smoothness, compatibility, and/or so forth to the updates across the frames. However, the second filter may apply a greater weight to the current weights **112** as compared to the previous weights **112** when there is a significant change in color associated with the current frame. For example, the second filter **120** may apply 90% (and/or any other percentage) to the current weights **112** and 10% (and/or any other percentage) to the previous weights **112** when there is a significant change in color. This way, the color processing performed on the current frame may be different from the color processing performed on the previous frame since the color (e.g., the brightness) between the frames is significantly different.

[0038] For instance, FIG. **2** illustrates an example of using a filter to determine weights associated with frames, in accordance with some embodiments of the present disclosure. As shown, over a period of time **202**, frames **204**(**1**)-(**5**) associated with an application may be processed. For instance, at a first time T(**1**), a first frame **204**(**1**) may be processed in order to determine first weights **206**(**1**) associated with the first frame **204**(**1**), such as by using the machine learning model(s) **108**. First applied weights **208**(**1**) to use when processing the first frame **204**(**1**) may then be determined using a weight **210** associated with the first weights **206**(**1**). In the example of FIG. **2**, the weight **210** may include 100% since the first frame **204**(**1**) is an initial frame for the processing.

[0039] Next, at a second time T(**2**), a second frame **204**(**2**) may be processed in order to determine second weights **206**(**2**) associated with the second frame **204**(**2**), such as by using the machine learning model(s) **108**. Second applied weights **208**(**2**) to use when processing the second frame **204**(**2**) may then be determined using a weight **212** associated with the second weights **206**(**2**) and a weight **214** associated with the first applied weights **208**(**1**). In the example of FIG. **2**, the weight **212** may include 10% and the weight **214** may include 90% since there is no significant change in color between the frames **204**(**1**)-(**2**). Next, at a third time T(**3**), a third frame **204**(**3**) may be processed in order to determine third weights **206**(**3**) associated with the third frame **204**(**3**), such as by using the machine learning model(s) **108**. Third applied weights **208**(**3**) to use when processing the third frame **204**(**3**) may then be determined using a weight **216** associated with the third weights **206**(**3**) and a weight **218** associated with the second applied weights **208**(**2**). In the example of FIG. **2**, the weight **216** may include 10% and the weight **218** may include 90% since there is no significant change in color between the frames **204**(**2**)-(**3**).

[0040] Next, at a fourth time T(**4**), a fourth frame **204**(**4**) may be processed in order to determine fourth weights **206**(**4**) associated with the fourth frame **204**(**4**), such as by using the machine learning model(s) **108**. Fourth applied weights **208**(**4**) to use when processing the fourth frame **204**(**4**) may then be determined using a weight **220** associated with the fourth weights **206**(**4**) and a weight **222** associated with the third applied weights **208**(**3**). In the example of FIG. **2**, the weight **220** may include 10% and the weight **222** may include 90% since there is no significant change in color between the frames **204**(**3**)-(**4**). Finally, at a fifth time T(**5**), a fifth frame **204**(**5**) may be processed in order to determine fifth weights **206**(**5**) associated with the fifth frame **204**(**5**), such as by using the machine learning model(s) **108**. Fifth applied weights **208**(**5**) to use when processing the fifth frame **204**(**5**) may be determined using a weight **224** associated with the fifth weights **206**(**5**) and a weight **226** associated with the fourth applied weights **208**(**4**). In the example of FIG. **2**, the weight **224** may include 90% and the weight **226** may include 10% since there is a significant change **228** in color between the frames **204**(**4**)-(**5**).

[0041] In the example of FIG. **2**, the determined weights **206**(**1**)-(**5**) and/or the applied weights **208**(**1**)-(**5**) may represent, and/or include, the weights **112**. Additionally, while the example of FIG. **2** describes using percentages that include 10% and 90%, in other examples, any other percentages may be used when performing the processes described herein.

[0042] Referring back to the example of FIG. **1**A, the process **100** may include a processing component **124** using the LUT **118** to transform a frame represented by the image data **104** into an updated frame represented by image data **126**. As described herein, the processing component **124** may use any technique to transform the frame to the updated frame using the LUT **118**. Additionally, the process **100** may continue to repeat such that one or more additional frames represented by the image data **104** are transformed into one or more updated frames represented by the image data **126**.

As described herein, by performing one or more of the processes described herein, the updated frame(s) may include on or more updated visual characteristics that cause the updated frame(s) to be associated with one or more reaction times that are less than one or more reaction times associated with the frame(s) before processing. Calculating reaction times associated with frames is described more herein with respect to at least the training (e.g., updating) of machine learning models.

[0043] While the example of FIG. **1**A describes generating the final LUT **118** using the LUTs **114**(**1**)-(**3**) and then using the final LUT **118** to transform the frame into the updated frame, in other examples, the process **100** may include directly applying the LUTs **114**(**1**)-(**3**) to the frame. For instance, the processing component **124** may directly apply the LUTs **114**(**1**)-(**3**) to the frame in order to transform the frame into the update frame represented by the image data **126**. When directly applying the LUTs **114**(**1**)-(**3**), the processing component **124** may use the weights **112**(**1**)-(**3**) and/or the processing component **124** may not use the weights **112**(**1**)-(**2**). In such examples, by performing such processes, the updated frame(s) may again include on or more updated visual characteristics that cause the updated frame(s) to be associated with one or more reaction times that are less than one or more reaction times associated with the frame(s) before processing.

[0044] FIG. **1**B illustrates another example of another process **128** of using one or more machine learning models to process frames in order to reduce reaction times associated with an application, in accordance with some embodiments of the present disclosure. As shown, image data **130** (which may be similar to, and/or include, the image data **104**) representing a frame of an application may be processed to generate processed image data **132** (which may be similar to, and/or include, the processed image data **106**) representing a processed frame. For instance, and as described herein, the frame represented by the image data **130** may include a first resolution while the processed frame represented by the image data **132** may include a second, lower resolution.

[0045] The processed image data **132** may then be processed using multiple layers of a machine learning model(s), such as multiple convolution layers **134**(**1**)-(**6**), multiple rectifier layers **136**(**1**)-(**5**), and multiple instance normalization layers **138**(**1**)-(**5**). While the example of FIG. **1**B illustrates a specific number of the convolution layers **134** (**1**)-(**6**), the rectifier layers **136**(**1**)-(**5**), and the instance normalization layers **138**(**1**)-(**5**), in other examples, the machine learning model(s) may include any number of convolution layers, rectifier layers, and/or instance normalization layers. Additionally, and as shown, the convolution layer **134**(**6**) may receive a color value **140** associated with the frame, where the color value **140** may be similar to and/or include the color value **122**. For example, the color value **140** may represent a mean color value associated with one or more frames being processed by the machine learning model(s). The convolution layer **134**(**6**) may then perform one or more of the processes described above with respect to the color component **116** to update one or more weights associated with one or more LUTs using the color value **140**.

[0046] The output from the convolution layer **134**(**6**) may then include a merged LUT **142**, which may be similar to and/or include the LUT **118**. Additionally, a processing component **144**, which may be similar to and/or include the processing component **124**, may apply the merged LUT **142** to the image data **130** in order to generate image data **146**, where the image data **146** may be similar to and/or include the image data **126**. For instance, the processing component **144** may apply the merged LUT **142** to the frame in order to generate an updated frame represented by the image data **146**. As described herein, by performing one or more of the processes described herein, the updated frame may include on or more updated visual characteristics that cause the updated frame to be associated with one or more reaction times that are less than one or more reaction times associated with the frame before processing. This process **128** may then continue to repeat for one or more additional frames represented by the image data **130**.

[0047] FIG. **1**C illustrates another example of another process **148** of using one or more machine learning models **150** to process frames in order to reduce reaction times associated with an application, in accordance with some embodiments of the present disclosure. The machine learning model(s) **150** may use the image data **152** (which may be similar to, and/or represent, the image data **104**), without or without preprocessing, as an input. As described herein, the image data **152** may represent one or more frames, such as one or more frames associated with an application. In some examples, the image data **152** may be input as a single frame, or may be input using batching, such as mini-batching. For example, two or more frames may be used as inputs together (e.g., at the same time).

[0048] The image data **152** may be input into one or more **154** layers (**1**)-(M) (also referred to singularly as "layer **154**" or in plural as "layers **154**") of the machine learning model (s) **150**. The layer(s) **154** may include any number of layers, such as one layer, six layers, fifty layers, and/or so forth. One or more of the layers **154** may include an input layer. The input layer may hold values associated with the image data **152**. For example, when the image data **152** represents a frame(s), the input layer may hold values representative of the raw pixel values of the frame(s) as a volume (e.g., a width, W, a height, H, and color channels, C (e.g., RGB), such as 32.times.32.times.3), and/or a batch size, B (e.g., where batching is used).

[0049] One or more layers **154** may include a convolutional layer. The convolutional layers may compute the output of neurons that are connected to local regions in an input layer (e.g., the input layer), each neuron computing a dot product between their weights and a small region they are connected to in the input volume. A result of a convolutional layer may be another volume, with one of the dimensions based on the number of filters applied (e.g., the width, the height, and the number of filters, such as 32×32×12, if 12 were the number of filters).

[0050] One or more of the layers **154** may include a rectified linear unit (ReLU) layer. The ReLU layer(s) may apply an elementwise activation function, such as the max (0, x), thresholding at zero, for example. The resulting volume of a ReLU layer may be the same as the volume of the input of the ReLU layer.

[0051] One or more of the layers **154** may include a pooling layer. The pooling layer may perform a down-sampling operation along the spatial dimensions (e.g., the height and the width), which may result in a smaller volume than the input of the pooling layer (e.g., 16×16×12 from the 32×32×12 input volume). In some examples, the machine learning model(s) **150** may not include any pooling layers.

In such examples, other types of convolution layers may be used in place of pooling layers. In some examples, the layer(s) **154** may include alternating convolutional layers and pooling layers.

[0052] One or more of the layers **154** may include a fully connected layer. Each neuron in the fully connected layer(s) may be connected to each of the neurons in the previous volume. The fully connected layer may compute class scores, and the resulting volume may be 1×1×N (where N is a number of classes). In some examples, the layer(s) **154** may include a fully connected layer, while in other examples, the fully connected layer of the machine learning model(s) **150** may be the fully connected layer separate from the layer(s) **154**. In some examples, no fully connected layers may be used by the layer(s) **154** and/or the machine learning model(s) **150** as a whole, in an effort to increase processing times and reduce computing resource requirements. In such examples, where no fully connected layers are used, the machine learning model(s) **150** may be referred to as a fully convolutional network.

[0053] One or more of the layers **154** may, in some examples, include a deconvolutional layer. However, the use of the term deconvolutional may be misleading and is not intended to be limiting. For example, the deconvolutional layer(s) may alternatively be referred to as transposed convolutional layers or fractionally strided convolutional layers. The deconvolutional layer(s) may be used to perform up-sampling on the output of a prior layer. For example, the deconvolutional layer(s) may be used to up-sample to a spatial resolution that is equal to the spatial resolution of the input frames to the machine learning model(s) **150**, or used to up-sample to the input spatial resolution of a next layer.

[0054] Although input layers, convolutional layers, pooling layers, ReLU layers, deconvolutional layers, and fully connected layers are discussed herein with respect to the layer(s) **154**, this is not intended to be limiting. For example, additional or alternative layers **154** may be used in the layer(s) **154**, such as normalization layers, SoftMax layers, and/or other layer types.

[0055] Different orders and numbers of the layers **154** of the machine learning model(s) **150** may be used, depending on the examples. In addition, some of the layers **154** may include parameters (e.g., weights and/or biases) while others may not, such as the ReLU layers and pooling layers, for example. In some examples, the parameters may be learned by the machine learning model(s) **150** during training, which is described herein. Further, some of the layers **154** may include additional hyper-parameters (e.g., learning rate, stride, epochs, kernel size, number of filters, type of pooling for pooling layers, etc.)—such as the convolutional layer(s), the deconvolutional layer(s), and the pooling layer(s)—while other layers may not, such as the ReLU layer(s). Various activation functions may be used, including but not limited to, ReLU, leaky ReLU, sigmoid, hyperbolic tangent (tan h), exponential linear unit (ELU), etc. The parameters, hyper-parameters, and/or activation functions are not to be limited and may differ depending on the embodiment.

[0056] In any example, the output of the machine learning model(s) **150** may include image data **156** (which may be similar to, and/or represent, the image data **126**). For instance, the image data **156** may represent one or more updated frames corresponding to the frame(s) represented by the image data **152**, where the updated frame(s) is associated with one or more reaction times that are less than one or more reaction times associated with the frame(s).

[0057] FIG. **3** illustrates an example of a process **300** of training the machine learning model(s) **108** to process frames in order to reduce reaction times associated with an application, in accordance with some embodiments of the present disclosure. As shown, the machine learning model(s) **108** may be trained using training image data **302**. As described herein, the training image data **302** may represent one or more training frames associated with one or more applications. In some examples, the training image data **302** may be specific to an application for which the machine learning model(s) **108** is being trained. In some examples, the training image data **302** may include general image data that is used to train the machine learning model(s) **108** for multiple applications. In any examples, and similar to the process **100**, the sampling component **102** may process the training image data **302** in order to generate processed image data **304**, using one or more of the processes described herein.

[0058] The process **300** may then include the machine learning model(s) **108** processing the processed image data **304**, using one or more of the processes described herein, in order to generate output data **306** (which may be similar to, and/or represent, the output data **110**) representing weights **308**(**1**)-(**3**) (which may be similar to, and/or represent, the weights **112**). Additionally, the process **300** may include the color component **116** performing one or more of the processes described herein to generate one or more LUTs **310** associated with the training input data **302** using at least the weights **308**(**1**)-(**3**) and the LUTs **114**. Furthermore, the process **300** may include the processing component **124** performing one or more of the processes described herein to generate image data **312** using the training input data **302** and the LUT(s) **310**, where the image data **312** represents one or more updated frames corresponding to the training frame(s) represented by the training image data **302**.

[0059] As shown, the process **300** may include a training engine **314** using the training image data **302**, the image data **312**, and ground truth data **316** to determine one or more losses **318** associated with the training image data **302**. As shown, the ground truth data **316** may include at least image data **320** representing one or more target frames corresponding to the training frame(s) represented by the training image data **302**. For example, for each training frame represented by the training image data **302**, the image data **320** may represent one or more corresponding target frames. As described herein, a target frame may be generated by updating one or more visual characteristics associated with a training frame. For instance, a target frame may be generated by updating one or more contrast levels, brightness levels, saturation levels, color levels, tone levels, and/or so forth associated with a training frame. In some examples, the visual characteristic(s) of the training frame may be updated to generate the target frame using one or more techniques. For example, the visual characteristic(s) may be updated in order to try and reduce a reaction time associated with the target frame as compared to the training frame. As described herein, in order to reduce the reaction time, the contrast levels, the brightness levels, and/or the saturation levels of the training frame may be increased to generate the target frame.

[0060] The training engine **314** may use one or more loss functions to determine a first loss **318** between a training frame and a target frame. As described herein, the first loss **318** may be associated with training the machine learning model(s) **108** for smoothness and/or monotonicity regularization. Additionally, any type of loss function may be used, such as cross entropy loss, mean squared error, mean absolute error, mean bias error, and/or other loss function types. Furthermore, the training engine **314** may use any technique to determine the first loss **318**, such as based on a comparison of visual characteristics (e.g., pixel values) of the target frame to visual characteristics (e.g., pixel values) of the training frame. The training engine **314** may then perform similar processes to determine one or more additional first losses **318** associated with one or more additional training frames represented by the training image data **302**.

[0061] As further illustrated by the example of FIG. **3**, the ground truth data **316** may include reaction time data **322** representing one or more reaction times associated with one or more target frames. In some examples, one or more reaction times may be determined for a single target frame, where a respective reaction time is associated with a portion (e.g., a pixel, a group of pixels, etc.) associated with the target frame. For instance, the following equations may be used to determine a reaction time associated with a portion of a target frame:

$$RS = \frac{C}{C \cdot (50 + 5f - 20 \cdot \log(lL)) + 10^{0.12+0.09f-0.42 \cdot log(L)}} \tag{1}$$

$$RT = \frac{1}{RS} \tag{2}$$

[0062] In equations (1) and (2), RS is the reaction speed, C is a contrast (e.g., a computed Michelson contrast, etc.), L is a luminance (e.g., a mean neighborhood luminance, etc.), f is a frequency, and RT is the reaction time (which is the inverse of reaction speed). In some examples, a respective reaction time may be computed for each point (e.g., each pixel) of a target frame. In some examples, reaction times may be computed for one or more (e.g., each) level of a pyramid, such as a Laplacian Pyramid.

[0063] For instance, FIG. **4** illustrates an example of determining reaction times associated with a frame, in accordance with some embodiments of the present disclosure. As shown, image data **402** representing a frame may be processed in order to generate luminance data **404** associated with the frame and/or contrast data **406** associated with the frame. In some examples, the luminance data **404** may represent a Luminance Pyramid that includes image data **408**(**1**)-(N) representing a number of frames, where the frames may be associated with different spatial frequencies. For instance, the image data **408**(**1**) may represent a first frame associated with first luminance values corresponding to a first spatial frequency, the image data **408**(**2**) may represent a second frame associated with second luminance values corresponding to a second spatial frequency, the image data **408**(**3**) may represent a third frame associated with third luminance values corresponding to a third spatial frequency, and/or so forth until the image data **408**(N) represents a last frame associated with last luminance values associated with a last spatial frequency.

[0064] Additionally, in some examples, the contrast data **406** may represent a Contrast Pyramid that includes image data **410**(**1**)-(N) representing a number of frames, where the frames may be associated with different spatial frequencies. For instance, the image data **410**(**1**) may represent a first frame associated with first contrast values (and/or first log (contrast) values) corresponding to a first spatial frequency, the image data **410**(**2**) may represent a second frame associated with second contrast values (and/or second log (contrast) values) corresponding to a second spatial frequency, the image data **408**(**3**) may represent a third frame associated with third contrast values (and/or third log (contrast) values) corresponding to a third spatial frequency, and/or so forth until the image data **408**(N) represents a last frame associated with last contrast values (and/or last log (contrast) values) corresponding to a last spatial frequency. In some examples, the spatial frequencies associated with the luminance data **404** are similar to the spatial frequencies associated with the contrast data **406**. In some examples, one or more of the spatial frequencies associated with the luminance data **404** may be different than one or more of the spatial frequencies associated with the contrast data **406**.

[0065] As described herein, the luminance data **404** and the contrast data **406** may be used to generate reaction time data **412** representing one or more reaction times associated with the frame. For example, the luminance data **404** may be used to determine the luminance values and the contrast data **406** may be used to determine the contrast values that are then used to determine the reaction times, such as using one or more of the equations (1)-(2). Additionally, in some examples, the reaction time data **412** may represent a respective reaction time associated with one or more (e.g., each) point (e.g., pixel, group of pixels, etc.) associated with the frame. Furthermore, in some examples, the reaction time data **412** may represent reaction times associated with one or more (e.g., each) level of the pyramids. Similar processes may then be used to generate additional reaction time data **412** representing one or more additional reaction times associated with one or more additional frames represented by the image data **402**.

[0066] Referring back to the example of FIG. **3**, and for a training frame, the training engine **314** may perform one or more of the processes described herein to determine one or more reaction times **324** associated with the updated frame that corresponds to the training frame. The training engine **314** may then determine a second loss **318** using the reaction time(s) associated with the updated frame and the reaction time(s) associated with the target frame. As described herein, the second loss **318** may be associated with training the machine learning model(s) **108** to compute weights **308**(**1**)-(**3**) for processing frames such that the updated frames are associated with lower reaction times. Additionally, any type of loss function may be used, such as cross entropy loss, mean squared error, mean absolute error, mean bias error, and/or other loss function types. The training engine **314** may then perform similar processes to determine one or more second losses **318** associated with one or more additional training frames represented by the training image data **302**.

[0067] As further shown by the example of FIG. **3**, and for a training frame, the training engine **314** may use one or more of the first loss **318** and/or the second loss **318** to update one or more parameters and/or weights of the machine learning model(s) **108** and/or update one or more of the LUTs **114**. In some examples, the training engine **314** may determine a final loss **318** using the first loss **318** and the second loss **318** and then use the total loss **318** to update the parameter(s) and/or weight(s) of the machine learning model(s) **108** and/or update the LUT(s) **114**. Additionally, in some examples, the training engine **314** may continue to perform these processes for one or more additional training frames represented by the training image data **302**.

[0068] Now referring to FIGS. **5** and **6**, each block of methods **500** and **600**, described herein, comprises a computing process that may be performed using any combination of hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory. The methods **500** and **600** may also be embodied as computer-usable instructions stored on computer storage media. The methods **500** and **600** may be provided by a standalone application, a service or hosted service (standalone or in combination with another hosted service), or a plug-in to another product, to name a few. In addition, the methods **500** and **600** are described, by way of example, with respect to FIG. **1** and FIG. **3**. However, these methods **500** and **600** may additionally or alternatively be executed by any one system, or any combination of systems, including, but not limited to, those described herein.

[0069] FIG. **5** illustrates a flow diagram showing a method **500** for using one or more machine learning models to process frames in order to reduce reaction times associated with an application, in accordance with some embodiments of the present disclosure. The method **500**, at block B**502**, may include generating first image data representative of one or more frames associated with an application, the one or more frames including one or more first visual characteristics associated with one or more first reaction times. For instance, one or more application servers (e.g., an application server **702**) may generate the image data **104** representative of the frame(s). As described herein, the application server(s) may generate the image data **104** during a session associated with the application and/or based at least on input data representing one or more inputs received using a client device (e.g., a client device **704**). Additionally, the frame(s) may include the first visual characteristic(s) associated with the first reaction time(s).

[0070] The method **500**, at block B**504**, may include generating, based at least on one or more machine learning models processing the first image data, second image data representative of one or more updated frames, the one or more updated frames including one or more second visual characteristics associated with one or more second reaction times. For instance, the machine learning model(s) **108** may process the image data **104** and, based at least on the processing, generate the image data **126** representing the updated frame(s). As described herein, the updated frame(s) includes the second visual characteristic(s) that are determined by updating the first visual characteristic(s). Additionally, the second visual characteristic(s) may be associated with the second reaction time(s), where the second reaction time(s) may be less than the first reaction time(s).

[0071] The method **500**, at block B**506**, may include causing an output associated with the one or more updated frames. For instance, the application server(s) may cause the outputting of the updated frame(s). As described herein, the application server(s) may cause the output by sending the image data **126** to the client device, where the client device then uses the image data **126** to present (e.g., render) the updated frame(s).

[0072] FIG. **6** illustrates a flow diagram showing a method **600** for training one or more machine learning models to process frames in order to reduce reaction times associated with an application, in accordance with some embodiments of the present disclosure. The method **600**, at block B**602**, may include obtaining training data representative of one or more training frames and ground truth data associated with one or more target frames corresponding to the one or more training frames. For instance, the training image data **302** representing the training frame(s) and the ground truth data **316** representing the data corresponding to the target frame(s) may be received. As described herein, the ground truth data **316** may include the image data **320** representing the target frame(s) and/or the reaction time data **322** representing the reaction time(s) associated with the target frame(s).

[0073] The method **600**, at block B**604**, may include generating, based at least on one or more machine learning models processing the training data, image data representative of one or more updated frames. For instance, the machine learning model(s) **108** may process the training image data **302** and, based at least on the processing, generate the image data **312** representing the updated frame(s). As described herein, the updated frame(s) includes the visual characteristic(s) that is determined by updating the visual characteristic(s) associated with the training frame(s). For example, the update frame(s) may be generated by updating the training frame(s) using one or more of the LUTs **114**.

[0074] The method **600**, at block B**606**, may include determining one or more losses based at least on one or more first reaction times associated with the one or more updated frames and one or more second reaction times associated with the one or more target frames. For instance, the training engine **314** may determine the loss(es) **318** based at least on the first reaction time(s) determined for the updated frame(s) and the second reaction time(s) determined for the target frame(s). As described herein, the training engine **314** may determine the loss(es) **318** by at least comparing the first reaction time(s) corresponding to one or more points of the frame to the second reaction time(s) corresponding to the same point(s). In other words, the training engine **314** may determine the loss(es) **318** associated with the different point(s) of the updated frame(s).

[0075] The method **600**, at block B**608**, may include updating, based at least on the one or more losses, one or more parameters associated with the one or more machine learning models. For instance, the training engine **314** may use the loss(es) **318** to update the parameter(s) of the machine learning model(s) **108**. As described herein, in some examples, the training engine **314** may further use the loss(es) **318** to update one or more of the LUTs **114**.

Example Content Streaming System

[0076] Now referring to FIG. **7**, FIG. **7** is an example system diagram for a content streaming system **700**, in accordance with some embodiments of the present disclosure. FIG. **7** includes application server(s) **702** (which may include similar components, features, and/or functionality to the example computing device **800** of FIG. **8**), client device(s) **704** (which may include similar components, features, and/or functionality to the example computing device **800** of FIG. **8**), and network(s) **706** (which may be similar to the network(s) described herein). In some embodiments of the present disclosure, the system **700** may be implemented. The application session may correspond to a game streaming application (e.g., NVIDIA GEFORCE NOW), a remote desktop application, a simulation application (e.g., autonomous or semi-autonomous vehicle simulation), computer aided design (CAD) applications, virtual reality (VR) and/or augmented reality (AR) streaming applications, deep learning applications, and/or other application types.

[0077] In the system **700**, for an application session, the client device(s) **704** may only receive input data in response to inputs to the input device(s), transmit the input data to the application server(s) **702**, receive encoded display data from the application server(s) **702**, and display the display data on the display **724**. As such, the more computationally intense computing and processing is offloaded to the application server(s) **702** (e.g., rendering—in particular ray or path tracing—for graphical output of the application session is executed by the GPU(s) of the game server(s) **702**). In other words, the application session is streamed to the client device(s) **704** from the application server(s) **702**, thereby reducing the requirements of the client device(s) **704** for graphics processing and rendering.

[0078] For example, with respect to an instantiation of an application session, a client device **704** may be displaying a frame of the application session on the display **724** based on receiving the display data from the application server(s) **702**. The client device **704** may receive an input to one of the input device(s) and generate input data in response. The client device **704** may transmit the input data to the application server(s) **702** via the communication interface **720** and over the network(s) **706** (e.g., the Internet), and the application server(s) **702** may receive the input data via the communication interface **718**. The CPU(s) may receive the input data, process the input data, and transmit data to the GPU(s) that causes the GPU(s) to generate a rendering of the application session. For example, the input data may be representative of a movement of a character of the user in a game session of a game application, firing a weapon, reloading, passing a ball, turning a vehicle, etc. The rendering component **712** may render the application session (e.g., representative of the result of the input data) and the render capture component **714** may capture the rendering of the application session as display data (e.g., as image data capturing the rendered frame of the application session). The rendering of the application session may include ray or path-traced lighting and/or shadow effects, computed using one or more parallel processing units-such as GPUs, which may further employ the use of one or more dedicated hardware accelerators or processing cores to perform ray or path-tracing techniques—of the application server(s) **702**. In some embodiments, one or more virtual machines (VMs)—e.g., including one or more virtual components, such as vGPUs, vCPUs, etc.—may be used by the application server (s) **702** to support the application sessions. The encoder **716** may then encode the display data to generate encoded display data and the encoded display data may be transmitted to the client device **704** over the network(s) **706** via the communication interface **718**. The client device **704** may receive the encoded display data via the communication interface **720** and the decoder **722** may decode the encoded display data to generate the display data. The client device **704** may then display the display data via the display **724**.

[0079] The systems and methods described herein may be used for a variety of purposes, by way of example and without limitation, for machine control, machine locomotion, machine driving, synthetic data generation, model training, perception, augmented reality, virtual reality, mixed reality, robotics, security and surveillance, simulation and digital twinning, autonomous or semi-autonomous machine applications, deep learning, environment simulation, data center processing, conversational AI, light transport simulation (e.g., ray-tracing, path tracing, etc.), collaborative content creation for 3D assets, cloud computing and/or any other suitable applications.

[0080] Disclosed embodiments may be comprised in a variety of different systems such as automotive systems (e.g., a control system for an autonomous or semi-autonomous machine, a perception system for an autonomous or semi-autonomous machine), systems implemented using a robot, aerial systems, medial systems, boating systems, smart area monitoring systems, systems for performing deep learning operations, systems for performing simulation operations, systems for performing digital twin operations, systems implemented using an edge device, systems incorporating one or more virtual machines (VMs), systems for performing synthetic data generation operations, systems implemented at least partially in a data center, systems for performing conversational AI operations, systems for performing light transport simulation, systems for performing collaborative content creation for 3D assets, systems implemented at least partially using cloud computing resources, and/or other types of systems.

Example Computing Device

[0081] FIG. **8** is a block diagram of an example computing device(s) **800** suitable for use in implementing some embodiments of the present disclosure. Computing device **800** may include an interconnect system **802** that directly or indirectly couples the following devices: memory **804**, one or more central processing units (CPUs) **806**, one or more graphics processing units (GPUs) **808**, a communication interface **810**, input/output (I/O) ports **812**, input/output components **814**, a power supply **816**, one or more presentation components **818** (e.g., display(s)), and one or more logic units **820**. In at least one embodiment, the computing device(s) **800** may comprise one or more virtual machines (VMs), and/or any of the components thereof may comprise virtual components (e.g., virtual hardware components). For non-limiting examples, one or more of the GPUs **808** may comprise one or more vGPUs, one or more of the CPUs **806** may comprise one or more vCPUs, and/or one or more of the logic units **820** may comprise one or more virtual logic units. As such, a computing device(s) **800** may include discrete components (e.g., a full GPU dedicated to the computing device **800**), virtual components (e.g., a portion of a GPU dedicated to the computing device **800**), or a combination thereof.

[0082] Although the various blocks of FIG. **8** are shown as connected via the interconnect system **802** with lines, this is not intended to be limiting and is for clarity only. For example, in some embodiments, a presentation component **818**, such as a display device, may be considered an I/O component **814** (e.g., if the display is a touch screen). As another example, the CPUs **806** and/or GPUs **808** may include memory (e.g., the memory **804** may be representative of a storage device in addition to the memory of the GPUs **808**, the CPUs **806**, and/or other components). In other words, the computing device of FIG. **8** is merely illustrative. Distinction is not made between such categories as "workstation," "server," "laptop," "desktop," "tablet," "client device," "mobile device," "hand-held device," "game console," "electronic control unit (ECU)," "virtual reality system," and/or other device or system types, as all are contemplated within the scope of the computing device of FIG. **8**.

[0083] The interconnect system **802** may represent one or more links or busses, such as an address bus, a data bus, a control bus, or a combination thereof. The interconnect system **802** may include one or more bus or link types, such as an industry standard architecture (ISA) bus, an extended industry standard architecture (EISA) bus, a video electronics standards association (VESA) bus, a peripheral component interconnect (PCI) bus, a peripheral component interconnect express (PCIe) bus, and/or another type of bus or link. In some embodiments, there are direct connections between components. As an example, the CPU **806** may be directly connected to the memory **804**. Further, the CPU **806** may be directly connected to the GPU **808**. Where there is direct, or point-to-point connection between components, the interconnect system **802** may include a PCIe link to carry out the connection. In these examples, a PCI bus need not be included in the computing device **800**.

[0084] The memory **804** may include any of a variety of computer-readable media. The computer-readable media may be any available media that may be accessed by the computing device **800**. The computer-readable media may include both volatile and nonvolatile media, and removable and non-removable media. By way of example, and not limitation, the computer-readable media may comprise computer-storage media and communication media.

[0085] The computer-storage media may include both volatile and nonvolatile media and/or removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, and/or other data types. For example, the memory **804** may store computer-readable instructions (e.g., that represent a program(s) and/or a program element(s), such as an operating system. Computer-storage media may include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which may be used to store the desired information and which may be accessed by computing device **800**. As used herein, computer storage media does not comprise signals per se.

[0086] The computer storage media may embody computer-readable instructions, data structures, program modules, and/or other data types in a modulated data signal such as a carrier wave or other transport mechanism and includes any information delivery media. The term "modulated data signal" may refer to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, the computer storage media may include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of any of the above should also be included within the scope of computer-readable media.

[0087] The CPU(s) **806** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **800** to perform one or more of the methods and/or processes described herein. The CPU(s) **806** may each include one or more cores (e.g., one, two, four, eight, twenty-eight, seventy-two, etc.) that are capable of handling a multitude of software threads simultaneously. The CPU(s) **806** may include any type of processor, and may include different types of processors depending on the type of computing device **800** implemented (e.g., processors with fewer cores for mobile devices and processors with more cores for servers). For example, depending on the type of computing device **800**, the processor may be an Advanced RISC Machines (ARM) processor implemented using Reduced Instruction Set Computing (RISC) or an x86 processor implemented using Complex Instruction Set Computing (CISC). The computing device **800** may include one or more CPUs **806** in addition to one or more microprocessors or supplementary co-processors, such as math co-processors.

[0088] In addition to or alternatively from the CPU(s) **806**, the GPU(s) **808** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **800** to perform one or more of the methods and/or processes described herein. One or more of the GPU(s) **808** may be an integrated GPU (e.g., with one or more of the CPU(s) **806** and/or one or more of the GPU(s) **808** may be a discrete GPU. In embodiments, one or more of the GPU(s) **808** may be a coprocessor of one or more of the CPU(s) **806**. The GPU(s) **808** may be used by the computing device **800** to render graphics (e.g., 3D graphics) or perform general purpose computations. For example, the GPU(s) **808** may be used for General-Purpose computing on GPUs (GPGPU). The GPU(s) **808** may include hundreds or thousands of cores that are capable of handling hundreds or thousands of software threads simultaneously. The GPU(s) **808** may generate pixel data for output images in response to rendering commands (e.g., rendering commands from the CPU(s) **806** received via a host interface). The GPU(s) **808** may include graphics memory, such as display memory, for storing pixel data or any other suitable data, such as GPGPU data. The display memory may be included as part of the memory **804**. The GPU(s) **808** may include two or more GPUs operating in parallel (e.g., via a link). The link may directly connect the GPUs (e.g., using NVLINK) or may connect the GPUs through a switch (e.g., using NVSwitch). When combined together, each GPU **808** may generate pixel data or GPGPU data for different portions of an output or for different outputs (e.g., a first GPU for a first image and a second GPU for a second image). Each GPU may include its own memory, or may share memory with other GPUs.

[0089] In addition to or alternatively from the CPU(s) **806** and/or the GPU(s) **808**, the logic unit(s) **820** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **800** to perform one or more of the methods and/or processes described herein. In embodiments, the CPU(s) **806**, the GPU(s) **808**, and/or the logic unit(s) **820** may discretely or jointly perform any combination of the methods, processes and/or portions thereof. One or more of the logic units **820** may be part of and/or integrated in one or more of the CPU(s) **806** and/or the GPU(s) **808** and/or one or more of the logic units **820** may be discrete components or otherwise external to the CPU(s) **806** and/or the GPU(s) **808**. In embodiments, one or more of the logic units **820** may be a coprocessor of one or more of the CPU(s) **806** and/or one or more of the GPU(s) **808**.

[0090] Examples of the logic unit(s) **820** include one or more processing cores and/or components thereof, such as Data Processing Units (DPUs), Tensor Cores (TCs), Tensor Processing Units (TPUs), Pixel Visual Cores (PVCs), Vision Processing Units (VPUs), Graphics Processing Clusters (GPCs), Texture Processing Clusters (TPCs), Streaming Multiprocessors (SMs), Tree Traversal Units (TTUs), Artificial Intelligence Accelerators (AIAs), Deep Learning Accelerators (DLAs), Arithmetic-Logic Units (ALUs), Application-Specific Integrated Circuits (ASICs), Floating Point Units (FPUs), input/output (I/O) elements, peripheral component interconnect (PCI) or peripheral component interconnect express (PCIe) elements, and/or the like.

[0091] The communication interface **810** may include one or more receivers, transmitters, and/or transceivers that enable the computing device **800** to communicate with other computing devices via an electronic communication network, included wired and/or wireless communications. The communication interface **810** may include components and functionality to enable communication over any of a number of different networks, such as wireless networks (e.g., Wi-Fi, Z-Wave, Bluetooth, Bluetooth LE, ZigBee, etc.), wired networks (e.g., communicating over Ethernet or InfiniBand), low-power wide-area networks (e.g., LoRaWAN, SigFox, etc.), and/or the Internet. In one or more embodiments, logic unit(s) **820** and/or communication interface **810** may include one or more data processing units (DPUs) to transmit data received over a network and/or through interconnect system **802** directly to (e.g., a memory of) one or more GPU(s) **808**.

[0092] The I/O ports **812** may enable the computing device **800** to be logically coupled to other devices including the I/O components **814**, the presentation component(s) **818**, and/or other components, some of which may be built in to (e.g., integrated in) the computing device **800**. Illustrative I/O components **814** include a microphone, mouse, keyboard, joystick, game pad, game controller, satellite dish, scanner, printer, wireless device, etc. The I/O components **814** may provide a natural user interface (NUI) that processes air gestures, voice, or other physiological inputs generated by a user. In some instances, inputs may be transmitted to an appropriate network element for further processing. An NUI may implement any combination of speech recognition, stylus recognition, facial recognition, biometric recognition, gesture recognition both on screen and adjacent to the screen, air gestures, head and eye tracking, and touch recognition (as described in more detail below) associated with a display of the computing device **800**. The computing device **800** may be include depth cameras, such as stereoscopic camera systems, infrared camera systems, RGB camera systems, touchscreen technology, and combinations of these, for gesture detection and recognition. Additionally, the computing device **800** may include accelerometers or gyroscopes (e.g., as part of an inertia measurement unit (IMU)) that enable detection of motion. In some examples, the output of the accelerometers or gyroscopes may be used by the computing device **800** to render immersive augmented reality or virtual reality.

[0093] The power supply **816** may include a hard-wired power supply, a battery power supply, or a combination thereof. The power supply **816** may provide power to the computing device **800** to enable the components of the computing device **800** to operate.

[0094] The presentation component(s) **818** may include a display (e.g., a monitor, a touch screen, a television screen, a heads-up-display (HUD), other display types, or a combination thereof), speakers, and/or other presentation components. The presentation component(s) **818** may receive data from other components (e.g., the GPU(s) **808**, the CPU(s) **806**, DPUs, etc.), and output the data (e.g., as an image, video, sound, etc.).

Example Data Center

[0095] FIG. **9** illustrates an example data center **900** that may be used in at least one embodiments of the present disclosure. The data center **900** may include a data center infrastructure layer **910**, a framework layer **920**, a software layer **930**, and/or an application layer **940**.

[0096] As shown in FIG. **9**, the data center infrastructure layer **910** may include a resource orchestrator **912**, grouped computing resources **914**, and node computing resources ("node C.R.s") **916**(**1**)-**916**(N), where "N" represents any whole, positive integer. In at least one embodiment, node C.R.s **916**(**1**)-**916**(N) may include, but are not limited to, any number of central processing units (CPUs) or other processors (including DPUs, accelerators, field programmable gate arrays (FPGAs), graphics processors or graphics processing units (GPUs), etc.), memory devices (e.g., dynamic read-only memory), storage devices (e.g., solid state or disk drives), network input/output (NW I/O) devices, network switches, virtual machines (VMs), power modules, and/or cooling modules, etc. In some embodiments, one or more node C.R.s from among node C.R.s **916**(**1**)-**916**(N) may correspond to a server having one or more of the above-mentioned computing resources. In addition, in some embodiments, the node C.R.s **916**(**1**)-**9161**(N) may include one or more virtual components, such as vGPUs, vCPUs, and/or the like, and/or one or more of the node C.R.s **916**(**1**)-**916**(N) may correspond to a virtual machine (VM).

[0097] In at least one embodiment, grouped computing resources **914** may include separate groupings of node C.R.s **916** housed within one or more racks (not shown), or many racks housed in data centers at various geographical locations (also not shown). Separate groupings of node C.R.s **916** within grouped computing resources **914** may include grouped compute, network, memory or storage resources that may be configured or allocated to support one or more workloads. In at least one embodiment, several node C.R.s **916** including CPUs, GPUs, DPUs, and/or other processors may be grouped within one or more racks to provide compute resources to support one or more workloads. The one or more racks may also include any number of power modules, cooling modules, and/or network switches, in any combination.

[0098] The resource orchestrator **912** may configure or otherwise control one or more node C.R.s **916**(**1**)-**916**(N) and/or grouped computing resources **914**. In at least one embodiment, resource orchestrator **912** may include a software design infrastructure (SDI) management entity for the data center **900**. The resource orchestrator **912** may include hardware, software, or some combination thereof.

[0099] In at least one embodiment, as shown in FIG. **9**, framework layer **920** may include a job scheduler **928**, a configuration manager **934**, a resource manager **936**, and/or a distributed file system **938**. The framework layer **920** may include a framework to support software **932** of software layer **930** and/or one or more application(s) **942** of application layer **940**. The software **932** or application(s) **942** may respectively include web-based service software or applications, such as those provided by Amazon Web Services, Google Cloud and Microsoft Azure. The framework layer **920** may be, but is not limited to, a type of free and open-source software web application framework such as Apache Spark™ (hereinafter "Spark") that may utilize distributed file system **938** for large-scale data processing (e.g., "big data"). In at least one embodiment, job scheduler **928** may include a Spark driver to facilitate scheduling of workloads supported by various layers of data center **900**. The configuration manager **934** may be capable of configuring different layers such as software layer **930** and framework layer **920** including Spark and distributed file system **938** for supporting large-scale data processing. The resource manager **936** may be capable of managing clustered or grouped computing resources mapped to or allocated for support of distributed file system **938** and job scheduler **928**. In at least one embodiment, clustered or grouped computing resources may include grouped computing resource **914** at data center infrastructure layer **910**. The resource manager **936** may coordinate with resource orchestrator **912** to manage these mapped or allocated computing resources.

[0100] In at least one embodiment, software **932** included in software layer **930** may include software used by at least portions of node C.R.s **916**(**1**)-**916**(N), grouped computing resources **914**, and/or distributed file system **938** of framework layer **920**. One or more types of software may include, but are not limited to, Internet web page search software, e-mail virus scan software, database software, and streaming video content software.

[0101] In at least one embodiment, application(s) **942** included in application layer **940** may include one or more types of applications used by at least portions of node C.R.s **916**(**1**)-**916**(N), grouped computing resources **914**, and/or distributed file system **938** of framework layer **920**. One or more types of applications may include, but are not limited to, any number of a genomics application, a cognitive compute, and a machine learning application, including training or inferencing software, machine learning framework software (e.g., PyTorch, TensorFlow, Caffe, etc.), and/or other machine learning applications used in conjunction with one or more embodiments.

[0102] In at least one embodiment, any of configuration manager **934**, resource manager **936**, and resource orchestrator **912** may implement any number and type of self-modifying actions based on any amount and type of data acquired in any technically feasible fashion. Self-modifying actions may relieve a data center operator of data center **900** from making possibly bad configuration decisions and possibly avoiding underutilized and/or poor performing portions of a data center.

[0103] The data center **900** may include tools, services, software or other resources to train one or more machine learning models or predict or infer information using one or more machine learning models according to one or more embodiments described herein. For example, a machine learning model(s) may be trained by calculating weight parameters according to a neural network architecture using software and/or computing resources described above with respect to the data center **900**. In at least one embodiment, trained or deployed machine learning models corresponding to one or more neural networks may be used to infer or predict information using resources described above with respect to the data center **900** by using weight parameters calculated through one or more training techniques, such as but not limited to those described herein.

[0104] In at least one embodiment, the data center **900** may use CPUs, application-specific integrated circuits (ASICs), GPUs, FPGAs, and/or other hardware (or virtual compute resources corresponding thereto) to perform training and/or inferencing using above-described resources. Moreover, one or more software and/or hardware resources described above may be configured as a service to allow users to train or performing inferencing of information, such as image recognition, speech recognition, or other artificial intelligence services.

Example Network Environments

[0105] Network environments suitable for use in implementing embodiments of the disclosure may include one or more client devices, servers, network attached storage (NAS), other backend devices, and/or other device types. The client devices, servers, and/or other device types (e.g., each device) may be implemented on one or more instances of the computing device(s) **800** of FIG. **8**—e.g., each device may include similar components, features, and/or functionality of the computing device(s) **800**. In addition, where backend devices (e.g., servers, NAS, etc.) are implemented, the backend devices may be included as part of a data center **900**, an example of which is described in more detail herein with respect to FIG. **9**.

[0106] Components of a network environment may communicate with each other via a network(s), which may be wired, wireless, or both. The network may include multiple networks, or a network of networks. By way of example, the network may include one or more Wide Area Networks (WANs), one or more Local Area Networks (LANs), one or more public networks such as the Internet and/or a public switched telephone network (PSTN), and/or one or more private networks. Where the network includes a wireless telecommunications network, components such as a base station, a communications tower, or even access points (as well as other components) may provide wireless connectivity.

[0107] Compatible network environments may include one or more peer-to-peer network environments—in which case a server may not be included in a network environment—and one or more client-server network environments—in which case one or more servers may be included in a network environment. In peer-to-peer network environments, functionality described herein with respect to a server(s) may be implemented on any number of client devices.

[0108] In at least one embodiment, a network environment may include one or more cloud-based network environments, a distributed computing environment, a combination thereof, etc. A cloud-based network environment may include a framework layer, a job scheduler, a resource manager, and a distributed file system implemented on one or more of servers, which may include one or more core network servers and/or edge servers. A framework layer may include a framework to support software of a software layer and/or one or more application(s) of an application layer. The software or application(s) may respectively include web-based service software or applications. In embodiments, one or more of the client devices may use the web-based service software or applications (e.g., by accessing the service software and/or applications via one or more application programming interfaces (APIs)). The framework layer may be, but is not limited to, a type of free and open-source software web application framework such as that may use a distributed file system for large-scale data processing (e.g., "big data").

[0109] A cloud-based network environment may provide cloud computing and/or cloud storage that carries out any combination of computing and/or data storage functions described herein (or one or more portions thereof). Any of these various functions may be distributed over multiple locations from central or core servers (e.g., of one or more data centers that may be distributed across a state, a region, a country, the globe, etc.). If a connection to a user (e.g., a client device) is relatively close to an edge server(s), a core server(s) may designate at least a portion of the functionality to the edge server(s). A cloud-based network environment may be private (e.g., limited to a single organization), may be public (e.g., available to many organizations), and/or a combination thereof (e.g., a hybrid cloud environment).

[0110] The client device(s) may include at least some of the components, features, and functionality of the example computing device(s) **800** described herein with respect to FIG. **8**. By way of example and not limitation, a client device may be embodied as a Personal Computer (PC), a laptop computer, a mobile device, a smartphone, a tablet computer, a smart watch, a wearable computer, a Personal Digital Assistant (PDA), an MP3 player, a virtual reality headset, a Global Positioning System (GPS) or device, a video player, a video camera, a surveillance device or system, a vehicle, a boat, a flying vessel, a virtual machine, a drone, a robot, a handheld communications device, a hospital device, a gaming device or system, an entertainment system, a vehicle computer system, an embedded system controller, a remote control, an appliance, a consumer electronic device, a workstation, an edge device, any combination of these delineated devices, or any other suitable device.

[0111] The disclosure may be described in the general context of computer code or machine-useable instructions, including computer-executable instructions such as program modules, being executed by a computer or other machine, such as a personal data assistant or other handheld device. Generally, program modules including routines, programs, objects, components, data structures, etc., refer to code that perform particular tasks or implement particular abstract data types. The disclosure may be practiced in a variety of system configurations, including hand-held devices, consumer electronics, general-purpose computers, more specialty computing devices, etc. The disclosure may also be practiced in distributed computing environments where tasks are performed by remote-processing devices that are linked through a communications network.

[0112] As used herein, a recitation of "and/or" with respect to two or more elements should be interpreted to mean only one element, or a combination of elements. For example, "element A, element B, and/or element C" may include only element A, only element B, only element C, element A and element B, element A and element C, element B and element C, or elements A, B, and C. In addition, "at least one of element A or element B" may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B. Further, "at least one of element A and element B" may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B.

[0113] The subject matter of the present disclosure is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this disclosure. Rather, the inventors have contemplated that the claimed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms "step" and/or "block" may be used herein to connote different elements of methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly described.

Example Paragraphs

[0114] A: A method comprising: applying, to one or more machine learning models, first image data representative of one or more frames associated with an interactive application, the one or more frames including one or more visual characteristics associated with one or more first reaction times; generating, based at least on the one or more machine learning models processing the first image data, second image data representative of one or more updated frames, the one or more updated frames including one or more updated visual characteristics associated with one or more second reaction times that are less than the one or more first reaction times; and causing an output of the one or more updated frames represented by the second image data.

[0115] B: The method of paragraph A, wherein: the one or more visual characteristics associated with the one or more first reaction times include at least one of one or more first luminance values or one or more first contrast values associated with one or more pixels of the one or more frames; and the one or more updated visual characteristics associated with the one or more second reaction times include at least one of one or more second luminance values or one or more second contrast values associated with the one or more frames.

[0116] C: The method of either paragraph A or paragraph B, wherein the generating the second image data comprises: determining, based at least on the one or more machine learning models processing the first image data, one or more weights associated with one or more first color lookup tables; and determining a second color lookup table based at least on the one or more weights and the one or more first color lookup tables; and generating the second image data by applying one or more values of the second color lookup table to one or more pixels of the one or more frames represented by the first image data.

[0117] D: The method of any one of paragraphs A-C, wherein the generating the second image data comprises: determining, based at least on the one or more machine learning models processing the first image data, one or more weights associated with one or more lookup tables; and generating the second image data by applying, based at least on the one or more weights, the one or more lookup tables to one or more pixels of the one or more frames represented by the first image data.

[0118] E: The method of any one of paragraphs A-D, further comprising: applying, to the one or more machine learning models, third image data representative of one or more second frames associated with the interactive application, the one or more second frames including one or more second visual characteristics associated with one or more third reaction times; generating, based at least on the one or more machine learning models processing the third image data, fourth image data representative of one or more updated second frames, the one or more updated second frames including one or more second updated visual characteristics associated with one or more fourth reaction times that are less than the one or more third reaction times; and causing an output of the one or more second updated frames represented by the fourth image data.

[0119] F: The method of paragraph E, wherein the generating the fourth image data comprises: determining, based at least on the one or more machine learning models processing the first image data, one or more first weights associated with one or more color lookup tables; determining, based at least on the one or more machine learning models processing the third image data, one or more second weights associated with the one or more color lookup tables; determining one or more third weights based at least on the one or more first weights, the one or more second weights, and the color value; and generating the fourth image data based at least on the third image data, the one or more third weights, and the one or more lookup tables.

[0120] G: The method of any one of paragraphs A-F, wherein the one or more machine learning models are updated, at least, by: generating, based at least on the one or more machine learning models processing third image data representative of one or more second frames, fourth image data representative of one or more updated second frames; determining one or more third reaction times associated with the one or more updated second frames; determining one or more fourth reaction times associated with one or more target frames corresponding to the one or more second frames; determining one or more loss values corresponding to one or more loss functions based at least on the one or more third reaction times and the one or more fourth reaction times; and updating, based at least on the one or more loss values, one or more parameters of the one or more machine learning models.

[0121] H: The method of paragraph G, wherein: the determining the one or more third reaction times is based at least on one or more second visual characteristics associated with the one or more second frames; and the determining the one or more fourth reaction times is based at least on one or more updated second visual characteristics associated with the one or more updated second frames.

[0122] I: The method of any one of paragraphs A-H, wherein the one or more machine learning models are updated, at least, by: generating, based at least on the one or more machine learning models processing third image data representative of one or more second frames, fourth image data representative of one or more updated second frames; determining one or more losses based at least on analyzing the one or more second frames with respect to one or more target frames; and updating, based at least on the one or more losses, one or more parameters of the one or more machine learning models.

[0123] J: The method of any one of paragraph A-I, wherein the causing the output of the one or more updated frames comprises one or more of: sending the second image data to one or more client devices for output by the one or more client devices; or displaying, using a client device, the one or more updated frames represented by the second image data.

[0124] K: A system comprising: one or more processors to: obtain first image data representative of one or more frames associated with an application, the one or more frames including one or more visual characteristics associated with one or more first reaction times; generate, based at least on one or more machine learning models processing the first image data, second image data representative of one or more updated frames, the one or more updated frames including one or more updated visual characteristics associated with one or more second reaction times that are less than the one or more first reaction times; and transmit the second image data to a client device.

[0125] L: The system of paragraph K, wherein: the one or more visual characteristics associated with the one or more first reaction times include at least one of one or more first luminance values or one or more first contrast values associated with one or more pixels of the one or more frames; and the one or more updated visual characteristics associated with the one or more second reaction times include at least one value of one or more second luminance values or one or more second contrast values associated with the one or more pixels.

[0126] M: The system of either paragraph K or paragraph L, wherein the generation of the second image data comprises: determining, based at least on the one or more machine learning models processing the first image data, one or more weights associated with one or more first color lookup tables; determining a second color lookup table based at least on the one or more weights and the one or more first color lookup tables; and generating the second image data by applying one or more values of the second color lookup table to one or more pixels of the one or more frames represented by the first image data.

[0127] N: The system of any one of paragraphs K-M, wherein the one or more processors are further to: obtain third image data representative of one or more second frames associated with the application, the one or more second frames including one or more second visual characteristics associated with one or more third reaction times; generate, based at least on the one or more machine learning models processing the third image data, fourth image data representative of one or more updated second frames, the one or more updated second frames including one or more updated second visual characteristics associated with one or more fourth reaction times that are less than the one or more third reaction times; and transmit the fourth image data to the client device.

[0128] O: The system of paragraph N, wherein the one or more processors are further to: determine, based at least on the one or more visual characteristics and the one or more second visual characteristics, a color value associated with the one or more second frames, wherein the generation of the fourth image data is further based at least on the color value.

[0129] P: The system of paragraph O, wherein the generation of the fourth image data comprises: determining, based at least on the one or more machine learning models processing the first image data, one or more first weights associated with one or more color lookup tables comprising one or more values corresponding to at least one visual characteristic of the first image data; determining, based at least on the one or more machine learning models processing the third image data, one or more second weights associated with the one or more color lookup tables; determining one or more third weights based at least on the one or more first weights, the one or more second weights, and the color value; and generating the fourth image data based at least on the third image data, the one or more third weights, and the one or more lookup tables.

[0130] Q: The system of any one paragraphs K-P, wherein the one or more machine learning models are updated, at least, by: generating, based at least on the one or more machine learning models processing third image data representative of one or more second frames, fourth image data representative of one or more updated second frames; determining one or more third reaction times associated with the one or more updated second frames; determining one or more fourth reaction times associated with one or more target frames corresponding to the one or more second frames; determining one or more loss values corresponding to one or more loss functions based at least on the one or more third reaction times and the one or more fourth reaction times; and updating, based at least on the one or more loss values, one or more parameters of the one or more machine learning models.

[0131] R: The system of any one of paragraphs K-Q, wherein the system is comprised in at least one of: a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for performing one or more simulation operations; a system for performing one or more digital twin operations; a system for performing light transport simulation; a system for performing collaborative content creation for 3D assets; a system for performing one or more deep learning operations; a system implemented using an edge device; a system implemented using a robot; a system for performing one or more generative AI operations; a system for performing operations using one or more large language models (LLMs); a system for performing operations using one or more vision language models (VLMs); a system for performing one or more conversational AI operations; a system for generating synthetic data; a system for presenting at least one of virtual reality content, augmented reality content, or mixed reality content; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.

[0132] S: One or more processors comprising: processing circuitry to generate, based at least on one or more machine learning models processing image data representative of one or more frames, updated image data representative of one or more updated frames that include one or more visual characteristics associated with a reaction time that is less than a reaction time associated with the one or more frames, and causing an output of the one or more updated frames represented by the updated image data.

[0133] T: The one or more processors of paragraph S, wherein the one or more processors are comprised in at least one of: a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for performing one or more simulation operations; a system for performing one or more digital twin operations; a system for performing light transport simulation; a system for performing collaborative content creation for 3D assets; a system for performing one or more deep learning operations; a system implemented using an edge device; a system implemented using a robot; a system for performing one or more generative AI operations; a system for performing operations using one or more large language models (LLMs); a system for performing operations using one or more vision language models (VLMs); a system for performing one or more conversational AI operations; a system for generating synthetic data; a system for presenting at least one of virtual reality content, augmented reality content, or mixed reality content; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.

What is claimed is:

1. A method comprising:

applying, to one or more machine learning models, first image data representative of one or more frames associated with an interactive application, the one or more frames including one or more visual characteristics associated with one or more first reaction times;

generating, based at least on the one or more machine learning models processing the first image data, second image data representative of one or more updated frames, the one or more updated frames including one or more updated visual characteristics associated with one or more second reaction times that are less than the one or more first reaction times; and

causing an output of the one or more updated frames represented by the second image data.

2. The method of claim **1**, wherein:

the one or more visual characteristics associated with the one or more first reaction times include at least one of one or more first luminance values or one or more first contrast values associated with one or more pixels of the one or more frames; and

the one or more updated visual characteristics associated with the one or more second reaction times include at least one of one or more second luminance values or one or more second contrast values associated with the one or more frames.

3. The method of claim **1**, wherein the generating the second image data comprises:

determining, based at least on the one or more machine learning models processing the first image data, one or more weights associated with one or more first color lookup tables; and

determining a second color lookup table based at least on the one or more weights and the one or more first color lookup tables; and

generating the second image data by applying one or more values of the second color lookup table to one or more pixels of the one or more frames represented by the first image data.

4. The method of claim **1**, wherein the generating the second image data comprises:

determining, based at least on the one or more machine learning models processing the first image data, one or more weights associated with one or more lookup tables; and

generating the second image data by applying, based at least on the one or more weights, the one or more lookup tables to one or more pixels of the one or more frames represented by the first image data.

5. The method of claim **1**, further comprising:

applying, to the one or more machine learning models, third image data representative of one or more second frames associated with the interactive application, the one or more second frames including one or more second visual characteristics associated with one or more third reaction times;

generating, based at least on the one or more machine learning models processing the third image data, fourth image data representative of one or more updated second frames, the one or more updated second frames including one or more second updated visual characteristics associated with one or more fourth reaction times that are less than the one or more third reaction times; and

causing an output of the one or more second updated frames represented by the fourth image data.

6. The method of claim **5**, wherein the generating the fourth image data comprises:

determining, based at least on the one or more machine learning models processing the first image data, one or more first weights associated with one or more color lookup tables;

determining, based at least on the one or more machine learning models processing the third image data, one or more second weights associated with the one or more color lookup tables;

determining one or more third weights based at least on the one or more first weights, the one or more second weights, and the color value; and

generating the fourth image data based at least on the third image data, the one or more third weights, and the one or more lookup tables.

7. The method of claim **1**, wherein the one or more machine learning models are updated, at least, by:

generating, based at least on the one or more machine learning models processing third image data representative of one or more second frames, fourth image data representative of one or more updated second frames;

determining one or more third reaction times associated with the one or more updated second frames;

determining one or more fourth reaction times associated with one or more target frames corresponding to the one or more second frames;

determining one or more loss values corresponding to one or more loss functions based at least on the one or more third reaction times and the one or more fourth reaction times; and

updating, based at least on the one or more loss values, one or more parameters of the one or more machine learning models.

8. The method of claim **7**, wherein:

the determining the one or more third reaction times is based at least on one or more second visual characteristics associated with the one or more second frames; and

the determining the one or more fourth reaction times is based at least on one or more updated second visual characteristics associated with the one or more updated second frames.

9. The method of claim **1**, wherein the one or more machine learning models are updated, at least, by:

generating, based at least on the one or more machine learning models processing third image data representative of one or more second frames, fourth image data representative of one or more updated second frames;

determining one or more losses based at least on analyzing the one or more second frames with respect to one or more target frames; and

updating, based at least on the one or more losses, one or more parameters of the one or more machine learning models.

10. The method of claim **1**, wherein the causing the output of the one or more updated frames comprises one or more of:

sending the second image data to one or more client devices for output by the one or more client devices; or

displaying, using a client device, the one or more updated frames represented by the second image data.

11. A system comprising:

one or more processors to:

obtain first image data representative of one or more frames associated with an application, the one or more frames including one or more visual characteristics associated with one or more first reaction times;

generate, based at least on one or more machine learning models processing the first image data, second image data representative of one or more updated frames, the one or more updated frames including one or more updated visual characteristics associated with one or more second reaction times that are less than the one or more first reaction times; and

transmit the second image data to a client device.

12. The system of claim **11**, wherein:

the one or more visual characteristics associated with the one or more first reaction times include at least one of one or more first luminance values or one or more first contrast values associated with one or more pixels of the one or more frames; and

the one or more updated visual characteristics associated with the one or more second reaction times include at least one value of one or more second luminance values or one or more second contrast values associated with the one or more pixels.

13. The system of claim **11**, wherein the generation of the second image data comprises:

determining, based at least on the one or more machine learning models processing the first image data, one or more weights associated with one or more first color lookup tables;

determining a second color lookup table based at least on the one or more weights and the one or more first color lookup tables; and

generating the second image data by applying one or more values of the second color lookup table to one or more pixels of the one or more frames represented by the first image data.

14. The system of claim **11**, wherein the one or more processors are further to:

obtain third image data representative of one or more second frames associated with the application, the one or more second frames including one or more second visual characteristics associated with one or more third reaction times;

generate, based at least on the one or more machine learning models processing the third image data, fourth image data representative of one or more updated second frames, the one or more updated second frames including one or more updated second visual characteristics associated with one or more fourth reaction times that are less than the one or more third reaction times; and

transmit the fourth image data to the client device.

15. The system of claim **14**, wherein the one or more processors are further to:

determine, based at least on the one or more visual characteristics and the one or more second visual characteristics, a color value associated with the one or more second frames,

wherein the generation of the fourth image data is further based at least on the color value.

16. The system of claim **15**, wherein the generation of the fourth image data comprises:

determining, based at least on the one or more machine learning models processing the first image data, one or more first weights associated with one or more color lookup tables comprising one or more values corresponding to at least one visual characteristic of the first image data;

determining, based at least on the one or more machine learning models processing the third image data, one or more second weights associated with the one or more color lookup tables;

determining one or more third weights based at least on the one or more first weights, the one or more second weights, and the color value; and

generating the fourth image data based at least on the third image data, the one or more third weights, and the one or more lookup tables.

17. The system of claim **11**, wherein the one or more machine learning models are updated, at least, by:

generating, based at least on the one or more machine learning models processing third image data representative of one or more second frames, fourth image data representative of one or more updated second frames;

determining one or more third reaction times associated with the one or more updated second frames;

determining one or more fourth reaction times associated with one or more target frames corresponding to the one or more second frames;

determining one or more loss values corresponding to one or more loss functions based at least on the one or more third reaction times and the one or more fourth reaction times; and

updating, based at least on the one or more loss values, one or more parameters of the one or more machine learning models.

18. The system of claim **11**, wherein the system is comprised in at least one of:

a control system for an autonomous or semi-autonomous machine;

a perception system for an autonomous or semi-autonomous machine;

a system for performing one or more simulation operations;

a system for performing one or more digital twin operations;

a system for performing light transport simulation;

a system for performing collaborative content creation for 3D assets;

a system for performing one or more deep learning operations;

a system implemented using an edge device;

a system implemented using a robot;

a system for performing one or more generative AI operations;

a system for performing operations using one or more large language models (LLMs);

a system for performing operations using one or more vision language models (VLMs);

a system for performing one or more conversational AI operations;

a system for generating synthetic data;

a system for presenting at least one of virtual reality content, augmented reality content, or mixed reality content;

a system incorporating one or more virtual machines (VMs);

a system implemented at least partially in a data center; or

a system implemented at least partially using cloud computing resources.

19. One or more processors comprising:

processing circuitry to generate, based at least on one or more machine learning models processing image data representative of one or more frames, updated image data representative of one or more updated frames that include one or more visual characteristics associated with a reaction time that is less than a reaction time associated with the one or more frames, and causing an output of the one or more updated frames represented by the updated image data.

20. The one or more processors of claim **19**, wherein the one or more processors are comprised in at least one of:

a control system for an autonomous or semi-autonomous machine;

a perception system for an autonomous or semi-autonomous machine;

a system for performing one or more simulation operations;

a system for performing one or more digital twin operations;

a system for performing light transport simulation;

a system for performing collaborative content creation for 3D assets;

a system for performing one or more deep learning operations;

a system implemented using an edge device;

a system implemented using a robot;

a system for performing one or more generative AI operations;

a system for performing operations using one or more large language models (LLMs);

a system for performing operations using one or more vision language models (VLMs);

a system for performing one or more conversational AI operations;

a system for generating synthetic data;

a system for presenting at least one of virtual reality content, augmented reality content, or mixed reality content;

a system incorporating one or more virtual machines (VMs);

a system implemented at least partially in a data center; or

a system implemented at least partially using cloud computing resources.

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